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Estimation of sleep problems among pregnant women during COVID-19 pandemic: A systematic review and meta-analysis

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Systematic review Article

Estimation of sleep problems among pregnant women during COVID-19 pandemic: A systematic review and meta-analysis

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Abstract

Objective: This systematic review and meta-analysis was conducted to estimate the sleep problems among pregnant women during the COVID-19 pandemic.

Methods: Five academic databases of Scopus, Medline/PubMed Central, ProQuest, ISI Web of Knowledge and Embase were searched systematically. Observational studies were included if sleep problem prevalence was assessed and reported using valid and reliable measure. There were no limitations regarding participants' characteristics. All the studies were published in English, peer-reviewed papers published between December 2019 and July 2021.

Results: A total of seven cross sectional studies with 2,808 participants from four countries were included. None of these papers gathered data during the national lockdown period in their respective countries. The pooled estimated prevalence of sleep problems was 56% [95% CI: 23 to 88%, I^2:99.81%, Tau^2:0.19]. Due to probability of publication bias, the fill-and-trim method was used to correct the estimated pooled measure. In this method, four studies were imputed and the corrected results based on this method showed that pooled prevalence of sleep problems was 13% (95% CI: 0 to 45%; p<.001). Based on meta-regression, age was the only significant predictor of prevalence of sleep problems among pregnant women, and accounted for 64% of the variance.

Conclusion: Pregnant women have experienced significant declines in sleep quality when faced with the COVID-19 pandemic. The short-term and long-term implications of such alterations in sleep on gestational and offspring outcomes are unclear and warrant further studies.

Keywords: COVID-19; Sleep problems; Pregnancy; meta-analysis

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Article summary

Strengths of this study

- The pooled prevalence of sleep problems was 13% (95% CI: 0 to 45%; p<.001) among pregnant women during covid pandemic.
- Age was the significant predictor of prevalence of sleep problems among pregnant women.
- Using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guideline and Newcastle Ottawa (NOS) checklist, relevant studies with satisfactory methodological quality searched from five academic databases were included and analyzed.

limitations of this study

- All studies were cross-sectional and COVID-19-related assessments at the time of data collection was not assessed.
- The outcomes of the pregnancies among those with and without sleep problems was not assessed...

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Introduction

Pregnant women are exposed to additional psychological stress due to the indirect adverse effects of COVID-19 pandemic [1]. The unintended consequences of COVID-19 can negatively impact the health of pregnant women and raise substantial concerns for this population [2].

Among these, mental health and psychological consequences, as well as sleep problems have become ubiquitously manifest across the world [3-6]. The United Nations has reported the COVID-19 pandemic as "the seeds of a major mental health crisis around the globe"[7]. In this context, pregnant women may be more vulnerable than others to the pandemic and its consequences and may be more at risk of COVID-19 infection and worse outcomes [8, 9]. Outbreaks of infectious diseases when combined with pregnancy are associated with major psychologic distress and significant symptoms, including poor sleep quality [10].

It is now well established that pregnancy in general is fraught with a higher risk of developing insomnia complaints, emergence of excessive daytime sleepiness and reports of decreased subjective sleep quality [11, 12]. Available studies estimated the prevalence of sleep problems such as insomnia and frequent awakenings in pregnant women at 46–78%, with the quality of sleep declining towards the third trimester[5]. Sleep disorders are associated with various adverse pregnancy-related health outcomes, including poorer fetal outcomes, specifically birth weight, growth, preterm birth and stillbirth [13]. Additionally, the occurrence of insomnia during pregnancy may increase the risk of postpartum depression [14].The result of a meta-analysis showed that the prevalence of anxiety and depression in pregnant women during COVID-19 pandemic was 15.8 and 25.8%, respectively [1].

The apparent associations between pregnancy and sleep disturbances, as well as the aforementioned adverse outcomes of COVID-19 in pregnant women raise the strong possibility that the changes in lifestyle and many other environmental circumstances imposed by the COVID-19 crisis may translate into enhanced repercussions on sleep during pregnancy. As such, we would anticipate that the prevalence of insomnia, a disorder associated with increased risk of chronic illnesses, poor mental health and functional limitations might be further enhanced [15]. It would be reasonable to assume that pregnant women may experience incremental sleep problems due to the social distancing and isolation required during the COVID-19 pandemic [16]. Thus, early diagnosis and timely treatment of such sleep problems may mitigate the risk for adverse gestational and perinatal outcomes [17].

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Despite the impact of the COVID-19 on sleep quality and quantity [18-20], little is known about how sleep among pregnant women has been affected by this current crisis. No systematic review or meta-analysis has yet been conducted to examine the impact of the pandemic on the prevalence of sleep problems during pregnancy. Therefore, this systematic review and meta-analysis was conducted to estimate the sleep problems among pregnant women during COVID-19 pandemic.

Methods

A systematic review was conducted through five academic databases. Relevant studies were extracted and their methodological quality was assessed using the Newcastle Ottawa Scale (NOS) checklist. Findings were synthesized using a meta-analysis approach. The report of present systematic review is part of a larger project registered in the PROSPERO International prospective register of systematic reviews (ID code: CRD42020181644)[21]. other related paper to this project is published elsewhere [22, 23]. This paper is prepared in accordance of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [24].

Search strategy

The five academic databases included Scopus, Medline/PubMed Central, ProQuest, ISI Web of Knowledge, and Embase which were searched systematically from December 2019 to July 2021. The search terms were extracted from PubMed Medical Subject Headings (MeSH). The main search terms were ‘sleep’, ‘COVID-19’ and ‘pregnancy’. The Boolean search method (AND/OR/NOT) was used to develop the search query. Search syntax was customized based on the advanced search attributes of each database.

Inclusion criteria

Observational studies were included if data on frequency or prevalence of sleep problems among pregnant women were reported. English, peer-reviewed papers published between December 2019 and July 2021 were included. There were no limitations regarding participant characteristics.

Outcomes

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Primary outcome: Estimates of sleep problems frequency was the primary outcome. Sleep problems required assessments using valid and reliable psychometric scales, or confirmed with defined cut-off points. For instance, a global score of 5 or more in the Pittsburgh Sleep Quality Index or a total score of 8 or more in the Insomnia Severity Index indicate poor sleep quality [25], [26].

Secondary outcomes: Assessing the possible sources of heterogeneity and predictor variables of sleep problems prevalence among pregnant individuals.

Study screening and selection

Screening of title and abstract were done independently by two researchers based on the inclusion criteria and any disagreements were resolved by consensus. The full texts of potentially relevant studies were further examined based on the aforementioned criteria. In this process, relevant studies were selected.

Quality assessment

The Newcastle Ottawa Scale (NOS) was used to evaluate the methodological quality of the studies in observational studies. Three characteristics (i.e., selection, comparability, and outcome) are examined with the NOS checklist. The checklist evaluates methodological quality of cross-sectional studies based on seven items. The maximum quality score is 9 points and studies with less than 5 points are classified as having a high risk of bias [27]. No studies were excluded based on the quality rating. However, the impact of quality on pooled effect size was assessed via meta-regression.

Data extraction

A pre-designed form was prepared to extract data from the studies included. The following items were extracted: first author's name, data collection dates, study design, country, number of participants, mean age, scales used to assess sleep problems, and numerical results regarding the frequency of sleep problems. It should also be noted that study selection, quality assessment, and data extraction were processes performed independently by two reviewers. Disagreements were resolved through discussion.

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Data synthesis

A quantitative synthesis using STATA software version 14 was conducted. Meta-analysis was run using a random effects model because included studies were taken from different populations, and both within-study and between-study variances should be accounted for [28]. Severity of heterogeneity was estimated using the I^2 index. Heterogeneity is interpreted as (i) mild when $I^2$is less than 25%, (ii)moderate when I^2 is 25 to50%, (iii) severe when I^2 is 50 to75%, and (iv) highly severe when $I^2$is greater than 75%[29]. The key measure selected for the present study was prevalence of sleep problems. The numerical findings regarding prevalence of sleep problems were reported consistently in 6 studies, and are reported along with 95% confidence intervals (CI). To assess moderator analysis, meta-regression was carried out. Funnel plot and Eggers' Test were used to assess publication bias [30].Meta-trim was used to correct probable publication bias [31]. The Jackknife method was used for sensitivity analysis [32].

Patient and Public Involvement: Patients or the public WERE NOT involved in the design, or conduct, or reporting, or dissemination plans of our research

Results

Study screening and selection process

The initial search in five databases resulted in 7,263 studies: Scopus (n=2,518), WOS (n=474), PubMed (n=338), Embase (n=1,426), and ProQuest (n=2,507). After removing duplicates, 5,647 papers retained based on title and abstract. Finally, 58 papers appeared to be potentially eligible and their full-texts were reviewed. In this process, 7 studies met the eligibility criteria, and were pooled in the meta-analysis. Figure 1 shows the search process based on the PRISMA flowchart.

Figure 1- PRISMA Flowchart of selected studies

Study description

A total of 7 papers with 2,808 participants from 4 different countries (Canada, Iran and Turkey (each 1 paper), and China (4 papers) were included. None of these papers gathered data during the national lockdown period in their respective countries. The smallest sample size was 45, and the largest sample size was 689, both from China. The mean age of participants was 30.66 years. All of the papers had cross-sectional design. Insomnia Severity Scale (ISI; n=4) and Pittsburgh

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Sleep Quality Index (PSQI; n=3) were used to assess sleep problems. Table 1 provides the summary characteristics of all included studies.

Quality assessment

All of the studies were categorized as being high quality studies. Recruitment of participants via online sampling and absence of estimate or justification regarding sample size were the most common problems encountered in the quality assessments.

Outcome measures

The pooled estimated prevalence of sleep problems was 56% [95% CI: 23 to 88%, I^2:99.81%, Tau^2:0.19]. Figure 2 provides the forest plot regarding the pooled prevalence.

Figure 2 - Forest plot regarding the pooled prevalence of sleep problems among pregnant women

The probability of publication bias was assessed using Egger's test and funnel plot. Based on Egger's test (p=0.03) and funnel plot (Figure 3), publication bias emerged as probable.

Figure 3- Funnel plot assessing the publication bias among included studies

Due to probability of publication bias, the fill-and-trim method was used to correct the estimated pooled measure. In this method, four studies were imputed, and the corrected results based on this method showed that the pooled prevalence of sleep problems was 13% (95% CI: 0 to 45%; p<.001). The resultant Funnel plot after trimming is provided in Figure 4.

Figure 4- Corrected Funnel plot based on fill and trim method

Furthermore, sensitivity analysis based on the Jack-knife method showed that the pooled effect size was not affected by a single study effect.

Figure 5- Assessment of small study effect based on Jack-knife method

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Meta-regression (Table 2) showed that none of the examined variables explained the observed heterogeneity. Age was the only significant predictor of prevalence of sleep problems among pregnant women, and accounted for 64% of the variance. Each year increase in participants' age was associated with a 12% decrease in the prevalence of sleep problems during pregnancy.

Discussion

The main objective of this systematic review and meta-analysis was to examine the prevalence of sleep problems among pregnant women during the COVID-19 pandemic, and explore potential predictors of such sleep problems. To the best of our knowledge, the present study is the first to summarize the available evidence on sleep problems, the latter being exclusively determined using validated instruments. We found that the overall prevalence of sleep problems during pregnancy in COVID-19 pandemic was nominally 56%, and after bias estimate corrections, 13%. Thus, sleep problems are extremely prevalent among pregnant women during the COVID-19 pandemic, and suggest the need to preventively address this issue to minimize the potential adverse consequences to the gestation itself or perinatal outcomes.

Pregnant women who excessively used electronic media were more likely to suffer from unhealthy lifestyles and poor sleep quality (38.2%). Zhang et al.(2021) in their study reported that poor sleep quality was associated with a higher risk of miscarriage during the COVID-19 pandemic [33].

Another study in China conducted by Zhou et al.(2020) found that the prevalence of insomnia symptoms in pregnancy during the COVID-19 pandemic was 2.6% [8]. These inordinately high prevalence figures raise the possibility that the gestational period is uniquely susceptible to sleep disruption by all of the environmental and lifestyle changes imposed by the pandemic control measures, even in the absence of infection.

The assumption was put forth that patients with COVID-19 would have the highest frequency of sleep disorders based on the fact that the core symptoms of the disease which involve coughing, fever, and difficulty in breathing may trigger concurrent sleep problems [34]. However, the determinants of sleep problems in the absence of COVID-19 but in the context of living through the pandemic have not been specifically delineated even though psychosocial stressors are clearly implicated [35-40]. Indeed, Targa et al.(2020) showed that COVID-19 outbreak correlate with declined sleep quality in association with an increase in negative mood [41], while Duran et

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al.(2021) findings showed that the COVID-19 pandemic had a negative effect on the psychological distress and sleep quality of adults[42]. Our findings in pregnant women are consistent with those of previous COVID-19 studies in the general population, and further buttress the perception that pregnancy generates a susceptible state that is prone to sleep disruption.

However, contrary findings have also been reported. For example, Zhou et al. (2020) conducted a survey of 544 pregnant women during the COVID-19 epidemic period in China, and found that only 2.6% suffered from insomnia [8]. Conversely, Xie et al.(2020) reported that 513 (74.5%) women (not necessarily pregnant) complained about sleep problems in the COVID-19 pandemic [43]. We posit that these discrepancies may reflect differences in the study population, measurement tools, and the timing and circumstances during which such surveys were performed.

Current findings indicate that the prevalence of sleep problems during pregnancy is particularly associated with maternal age and coexistence of the pregnancy with concurrent COVID-19 disease. Stress, negative emotions and repetitive negative thinking are combined with problems in initiating and maintaining sleep [44]. Worsened sleep may amplify the negative effects, resulting in increased susceptibility to mood and anxiety problems [45, 46]. Thus, pregnancy during a pandemic has implications for sleep regulation and wellness, and suggests the need for developing psychotherapeutic strategies to target domains.

Limitations

Among the limitations of the study we should point out that all studies were cross-sectional, absence of COVID-19-related assessments at the time of data collection, and the outcomes of the pregnancies among those with and without sleep problems in a consistent manner. However, these results support the need for targeted studies in cohorts of pregnant women aimed at early detection of vulnerable sub-groups along with initial behavioral interventions aimed at mitigating the potential negative consequences of living through a pandemic on sleep during pregnancy and downstream outcomes.

Conclusion:

Pregnant women are likely to experience poor sleep quality when facing a pandemic such as COVID-19. Thus, special attention should be paid to pregnant women, and tools to identify those

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at risk for sleep problems along with effective interventions to prevent or mitigate the consequences of sleep problems during gestation are urgently needed.

Authors' contributions: ZA & AHP contributed to the conception, design of the study, and data collection. ZA & AHP contributed in data analysing and interpretation of data. ZA and FA drafted the manuscript. AHP & DG provided contributions to the literature review and discussion and substantially edited the primary manuscript and prepared the final version of the manuscript. All authors revised the manuscript, agreed to be fully accountable for ensuring the integrity and accuracy of the study, and read and approved the final version of the manuscript to be published. All the authors met the criteria for authorship, and they are listed as co-authors on the title page.

Acknowledgments: None to declare

Conflict of interests: None to declare

Ethics, approval, and consent to participate: The current research protocol was registered in PROSPERO International prospective register of systematic reviews (ID code: CRD42020181644.

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Clinical Trial Registry and Registration number: Not applicable. Current study is systematic review and registered in PROSPERO International prospective register of systematic reviews.

Data sharing statement: No additional unpublished data are available.

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Table 1- Summary of the characteristics of included studies

Authors	Year	Country	Collection Date	Sample Size	Mean Age (Years)	NOS	Sleep Problem Scale
Li [8]	2020	China	25 April to 9 May 2020	398		9	ISI
Khoury [47]	2021	Canada	June 3 and July 31, 2020	303	32.13	7	ISI
Xie [43]	2020	China	-	689	29.03	6	PSQI
Zhang [33]	2021	China	January to February 2020	456		6	PSQI
Zhou [8]	2020	China	February 28–March 12, 2020	859	33.25	9	ISI
Ahorsu [48]	2020	Iran	March 7 and April 21 2020	290	29.24	9	ISI
Alan [49]	2020	Turkey	25-30 April 2020	166	29.65	10	PSQI

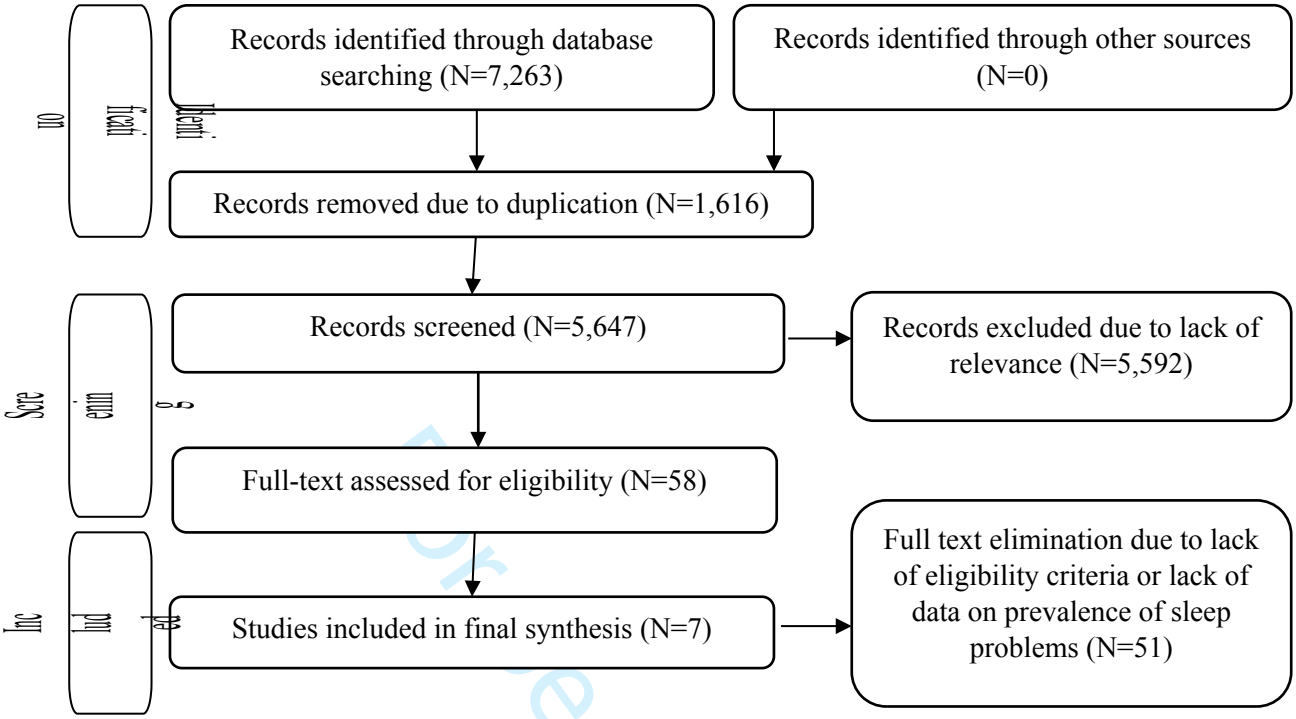
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Table 2. Results of meta-regression regarding estimated pooled prevalence						
univariable meta-regression						
Variable	Coefficient	S.E.	*p*	I^2 res. (%)	Adj. R^2 (%)	Tau^2
Country	0.15	0.08	0.11	99.72	32.23	0.05
Age	-0.15	0.05	0.07	98.76	64.01	0.04
NOS- score	0.02	0.06	0.74	99.82	-17.10	0.09
Measure of sleep	0.20	0.22	0.41	99.61	-3.06	0.09
univariable meta-regression						
Country	0.08	0.09	0.42	98.61	63.91	0.04
Age	-0.12	0.06	0.17			

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Figure 1- PRISMA Flowchart of selected studies



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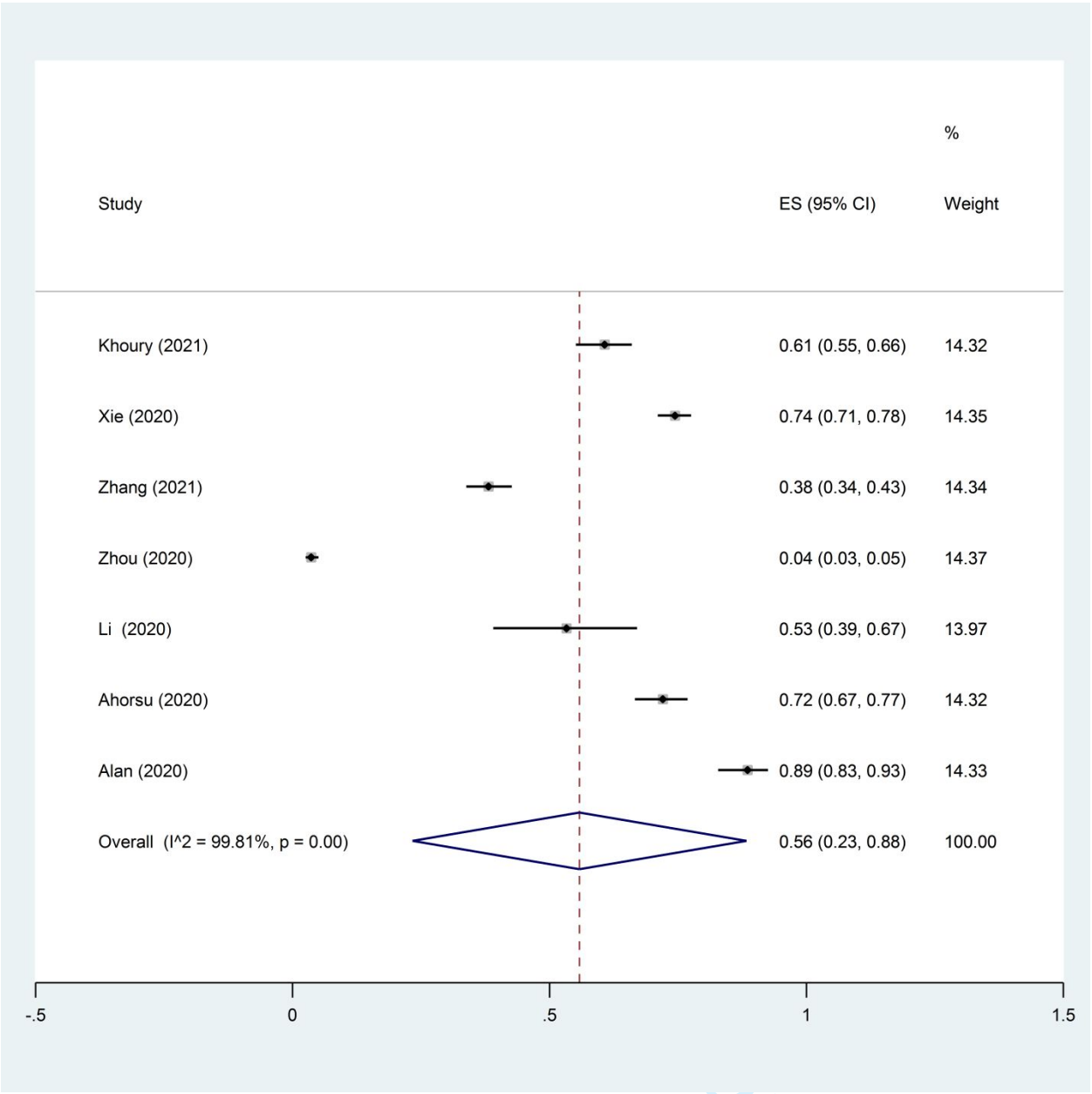


Figure 2 - Forest plot regarding the pooled prevalence of sleep problems among pregnant women

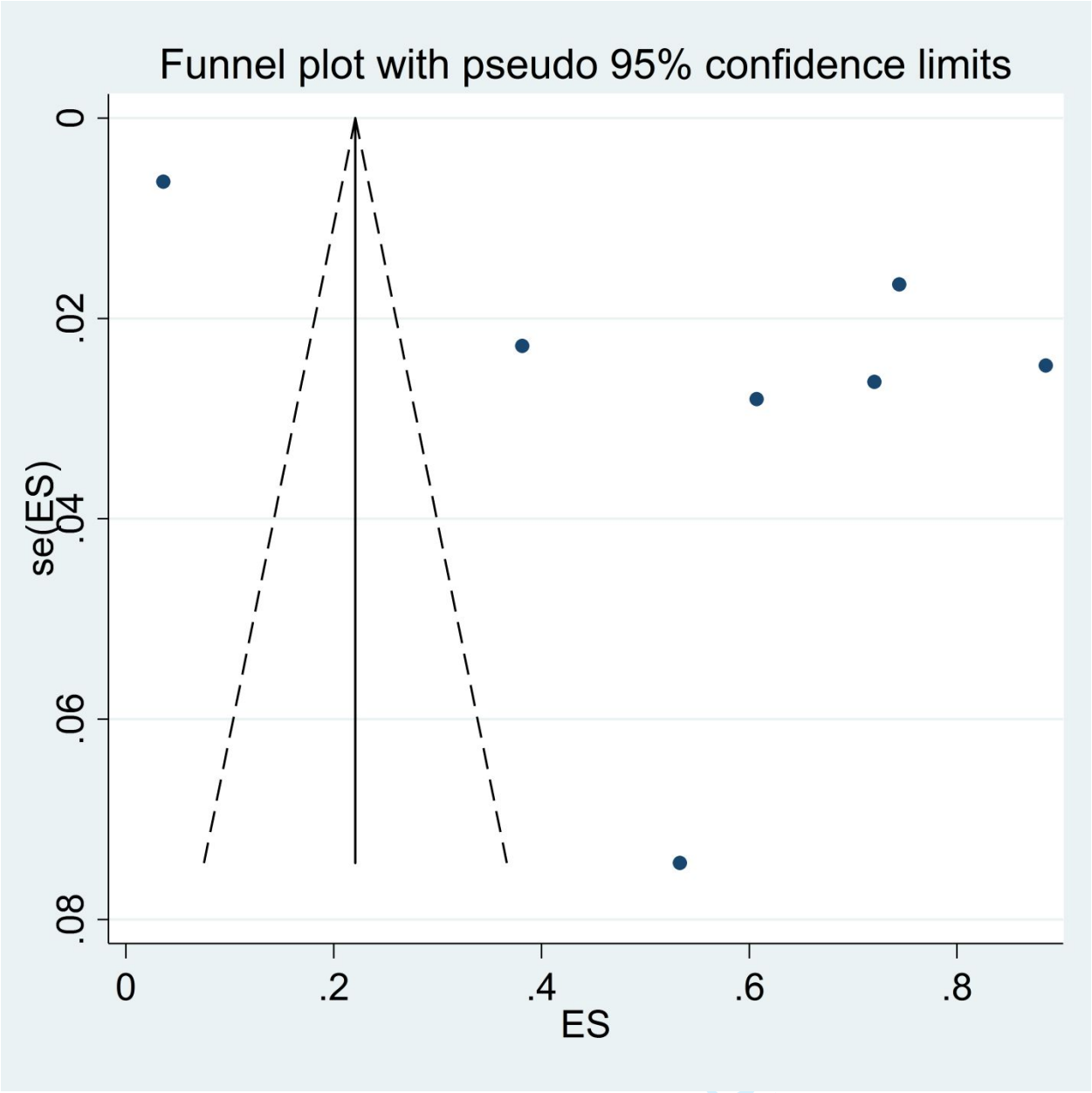


Figure 3- Funnel plot assessing the publication bias among included studies

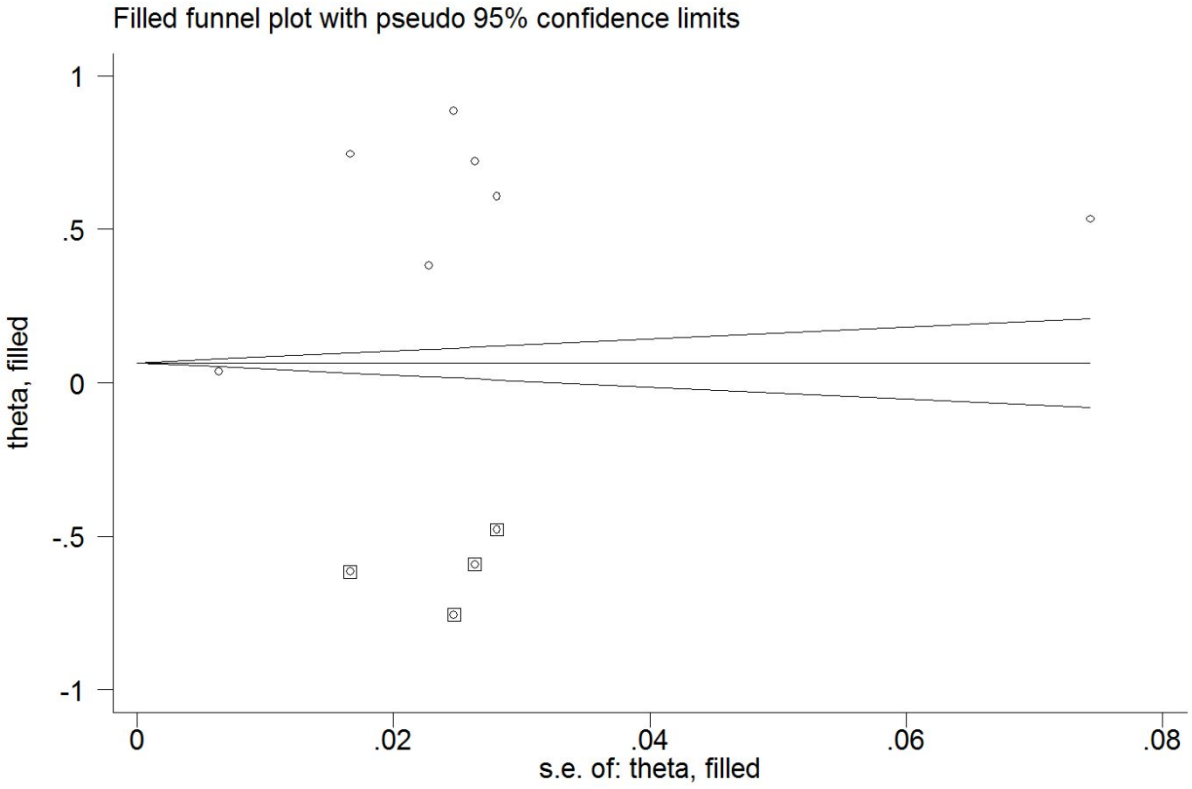


Figure 4- Corrected funnel plot based on fill and trim method

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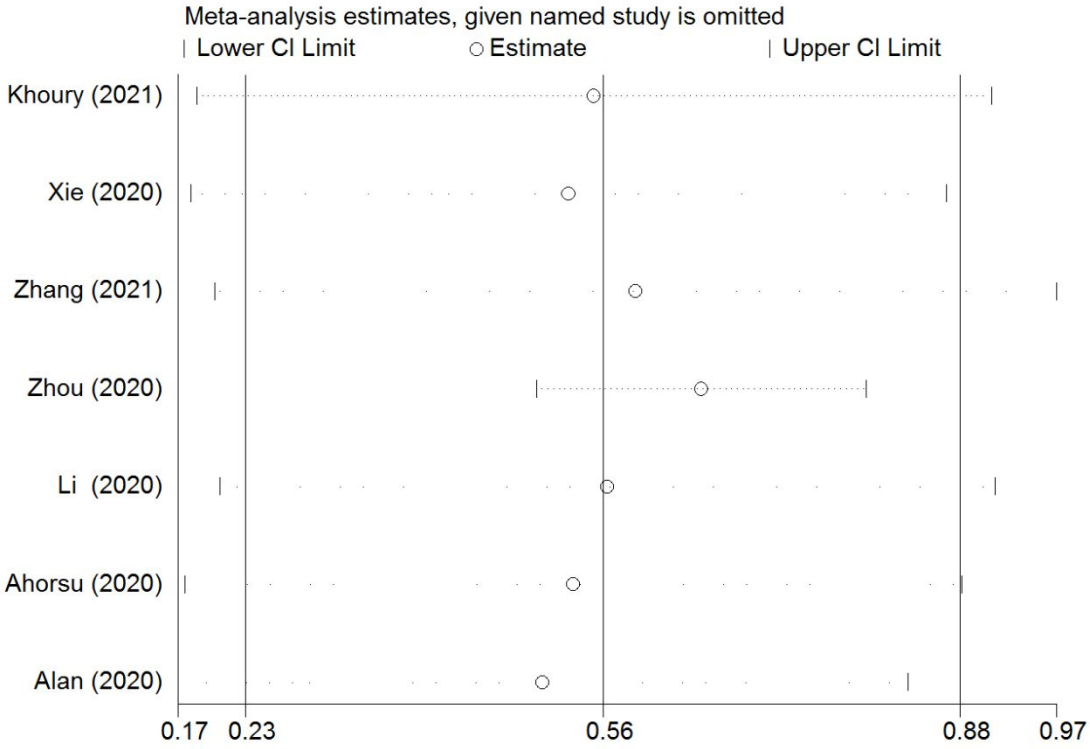


Figure 5- Assessment of small study effect based on Jack-knife method

view only

Database	Syntax
PubMed central	("2019 nCoV" OR 2019nCoV OR "2019 novel coronavirus" OR "COVID 19" OR COVID19 OR "new coronavirus" OR "novel coronavirus" OR "SARS CoV-2" OR (Wuhan AND coronavirus) OR "COVID 19" OR "SARS-CoV" OR "2019-nCoV" OR "SARS-CoV-2") AND (("insomnia" OR "sleep problems" OR "Sleep quality" OR Dyssomnia OR ("Sleep Disorders" AND Extrinsic) OR "Extrinsic Sleep Disorder" OR "Extrinsic Sleep Disorders" OR ("Sleep Disorder" AND Extrinsic) OR "Limit-Setting Sleep Disorder" OR "Limit Setting Sleep Disorder" OR "Limit-Setting Sleep Disorders" OR ("Sleep Disorders" AND "Limit-Setting") OR ("Sleep Disorder" AND Limit-Setting) OR ("Sleep Disorder" AND "Limit Setting") OR "Nocturnal Eating-Drinking Syndrome" OR ("Eating-Drinking Syndrome" AND Nocturnal) OR ("Eating-Drinking Syndromes" AND Nocturnal) OR "Nocturnal Eating Drinking Syndrome" OR "Nocturnal Eating-Drinking Syndromes" OR (Syndrome AND "Nocturnal Eating-Drinking") OR (Syndromes AND "Nocturnal Eating-Drinking") OR "Adjustment Sleep Disorder" OR "Adjustment Sleep Disorders" OR ("Sleep Disorders" AND Adjustment) OR ("Sleep Disorder" AND Adjustment) OR "Environmental Sleep Disorder" OR "Environmental Sleep Disorders" AND ("Sleep Disorders" AND Environmental) OR ("Sleep Disorder" AND Environmental)) AND ("2019/11/01"[PubDate] : "2020/12/31"[PubDate])
ProQuest	("2019 nCoV" OR 2019nCoV OR "2019 novel coronavirus" OR "COVID 19" OR COVID19 OR "new coronavirus" OR "novel coronavirus" OR "SARS CoV-2" OR (Wuhan AND coronavirus) OR "COVID 19" OR "SARS-CoV" OR "2019-nCoV" OR "SARS-CoV-2") AND (insomnia OR "sleep problems" OR "Sleep quality" OR Dyssomnia OR "Sleep Disorders" OR "Extrinsic Sleep Disorder" OR "Extrinsic Sleep Disorders" OR "Limit-Setting Sleep Disorder" OR "Limit Setting Sleep Disorder" OR "Limit-Setting Sleep Disorders" OR "Nocturnal Eating-Drinking Syndrome" OR ("Eating-Drinking Syndrome" AND Nocturnal) OR ("Eating-Drinking Syndromes" AND Nocturnal) OR "Nocturnal Eating Drinking Syndrome" OR "Nocturnal Eating-Drinking Syndromes" OR (Syndrome AND "Nocturnal Eating-Drinking") OR (Syndromes AND "Nocturnal Eating-Drinking") OR "Adjustment Sleep Disorder" OR "Adjustment Sleep Disorders" OR ("Sleep Disorders" AND Adjustment) OR ("Sleep Disorder" AND Adjustment) OR "Environmental Sleep Disorder" OR "Environmental Sleep Disorders" AND ("Sleep Disorders" AND Environmental)) OR ("Sleep Disorder" AND Environmental) AND ("2019/11/01"[PubDate] : "2020/12/31"[PubDate])
Scopus	("2019 nCoV" OR 2019ncov OR "2019 novel coronavirus" OR "COVID 19" OR covid19 OR "new coronavirus" OR "novel coronavirus" OR "SARS CoV-2" OR (wuhan AND coronavirus) OR "COVID 19" OR "SARS-CoV" OR "2019-nCoV" OR "SARS-CoV-2") AND (insomnia OR "sleep problems" OR "Sleep quality" OR dyssomnia OR "Sleep Disorders")
Embase	('2019 ncov' OR 2019ncov OR '2019 novel coronavirus'/exp OR '2019 novel coronavirus' OR covid19 OR 'new coronavirus' OR 'novel coronavirus' OR 'sars cov-2' OR (wuhan AND ('coronavirus'/exp OR coronavirus)) OR 'covid 19'/exp OR 'covid 19' OR 'sars-cov'/exp OR 'sars-cov' OR '2019-ncov' OR 'sars-cov-2') AND ('insomnia'/exp OR insomnia OR 'sleep problems' OR 'sleep quality'/exp OR 'sleep quality' OR 'dyssomnia'/exp OR dyssomnia OR 'sleep disorders'/exp OR 'sleep disorders')
ISI web of Knowledge	(TS= "2019 nCoV" OR TS= “2019ncov” OR TS= "2019 novel coronavirus" OR TS="COVID 19" OR TS=covid19 OR TS="new coronavirus" OR TS= "novel coronavirus" OR TS="SARS CoV-2" OR TS= (wuhan AND coronavirus) OR TS="COVID 19" OR TS="SARS-CoV" OR TS="2019-nCoV" OR TS="SARS-CoV-2") AND (TS=insomnia OR TS="sleep problems" OR TS="Sleep quality" OR TS= dyssomnia OR TS="Sleep Disorders")

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	1
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	2
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	3
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	3
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	3
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Appendix 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	4
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	4
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	4
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	4
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	4
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	5
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	5
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	NA
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	5
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	5
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	5
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	5
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	5
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	5

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	6
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	6
Study characteristics	17	Cite each included study and present its characteristics.	6
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	6
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table 1
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	6
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Figures 1-4 Table 2
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	7
	23b	Discuss any limitations of the evidence included in the review.	8
	23c	Discuss any limitations of the review processes used.	8
	23d	Discuss implications of the results for practice, policy, and future research.	9
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	3
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	3
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	10
Competing interests	26	Declare any competing interests of review authors.	10
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	10

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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Estimation of sleep problems among pregnant women during COVID-19 pandemic: A systematic review and meta-analysis

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Keywords:	SLEEP MEDICINE, OBSTETRICS, Maternal medicine < OBSTETRICS

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Systematic review Article

Estimation of sleep problems among pregnant women during COVID-19 pandemic: A systematic review and meta-analysis

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Abstract

Objective: *T*o estimate the sleep problems among pregnant women during the COVID-19 pandemic.

Eligibility criteria: English, peer-reviewed, observational studies published between December 2019 and July 2021 assessed and reported sleep problem prevalence using a valid and reliable measure were included

Information sources: Scopus, Medline/PubMed Central, ProQuest, ISI Web of Knowledge and Embase

Risk of bias assessment tool: The Newcastle Ottawa Scale (NOS) checklist

Synthesis of results: Prevalence of sleep problems was synthesis using STATA software version 14 using a random effects model. To assess moderator analysis, meta-regression was carried out. Funnel plot and Eggers' Test were used to assess publication bias. Meta-trim was used to correct probable publication bias. The Jackknife method was used for sensitivity analysis

Included studies: A total of seven cross sectional studies with 2,808 participants from four countries were included.

Synthesis of results: The pooled estimated prevalence of sleep problems was 56% [95% CI: 23 to 88%, I^2:99.81%, Tau^2:0.19]. Due to the probability of publication bias, the fill-and-trim method was used to correct the estimated pooled measure, which imputed four studies. The corrected results based on this method showed that pooled prevalence of sleep problems was 13% (95% CI: 0 to 45%; p<.001). Based on meta-regression, age was the only significant predictor of prevalence of sleep problems among pregnant women.

Limitations of evidence: all studies were cross-sectional, absence of assessment of sleep problems prior to COVID-19, and the outcomes of the pregnancies among those with and without sleep problems in a consistent manner are among limitation of current review.

Interpretation: Pregnant women have experienced significant declines in sleep quality when faced with the COVID-19 pandemic. The short-term and long-term implications of such alterations in sleep on gestational and offspring outcomes are unclear and warrant further studies.

Funding: Qazvin university of Medical Sciences

Registration: Registered in the PROSPERO with ID code of CRD42020181644.

Keywords: COVID-19; Sleep problems; Pregnancy; meta-analysis

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Article summary

Strengths of this study

- Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guideline was used to report the findings.
- Newcastle Ottawa (NOS) checklist was used to assess methodological quality of included studies.
- Five academic databases were systematically searched to increase the comprehensiveness of search.

limitations of this study

- All retrieved studies were cross-sectional.
- The outcomes of the pregnancies among those with and without sleep problems was not assessed in included studies.

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Introduction

Pregnant women are exposed to additional psychological stress due to the indirect adverse effects of COVID-19 pandemic [1, 2]. The unintended consequences of COVID-19 can negatively impact the health of pregnant women and raise substantial concerns for this population [3].

Among these, mental health and psychological consequences, as well as sleep problems have become ubiquitously manifest across the world [4-7]. On the other hand, people who have a good quality of sleep have fewer post-traumatic stress disorder (PTSD) symptoms associated with the COVID-19 outbreak [8, 9].The United Nations has reported the COVID-19 pandemic as "the seeds of a major mental health crisis around the globe"[10]. In this context, pregnant women may often develop psychiatric symptoms, reduced sleep quality, which may enhance their vulnerability relative to others to the pandemic and its consequences and may be more at risk of COVID-19 infection and worse outcomes [11, 12]. Outbreaks of infectious diseases when combined with pregnancy are associated with major psychologic distress and significant symptoms, including poor sleep quality [13].

It is now well established that pregnancy in general is fraught with a higher risk of developing insomnia complaints, emergence of excessive daytime sleepiness and reports of decreased subjective sleep quality [14, 15]. Available studies estimated the prevalence of sleep problems such as insomnia and frequent awakenings in pregnant women at 46–78%, with the quality of sleep declining towards the third trimester[6]. Sleep disorders are associated with various adverse pregnancy-related health outcomes, including poorer fetal outcomes, specifically birthweight, growth, preterm birth and stillbirth [16]. Additionally, the occurrence of insomnia during pregnancy may increase the risk of postpartum depression [17].The result of a meta-analysis showed that the prevalence of anxiety and depression in pregnant women during COVID-19 pandemic was 15.8 and 25.8%, respectively [1].

Pregnant women may be more likely to develop anxiety, the latter being powerfully linked with poor sleep, particularly during the COVID-19 pandemic [18]. The apparent associations between pregnancy and sleep disturbances, as well as the aforementioned adverse outcomes of COVID-19 in pregnant women raise the strong possibility that the changes in lifestyle and many other environmental circumstances imposed by the COVID-19 crisis may translate into enhanced repercussions on sleep during pregnancy. As such, we would anticipate that the prevalence of insomnia, a disorder associated with increased risk of chronic illnesses, poor mental health and

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functional limitations might be further enhanced [19]. It would be reasonable to assume that pregnant women may experience incremental sleep problems due to the social distancing and isolation required during the COVID-19 pandemic [20]. Thus, early diagnosis and timely treatment of such sleep problems may mitigate the risk for adverse gestational and perinatal outcomes [21]. Despite the impact of the COVID-19 on sleep quality and quantity [22-26], there is no systematic review or meta-analysis has yet been conducted to examine the impact of the pandemic on the prevalence of sleep problems during pregnancy. Therefore, this systematic review and meta-analysis was conducted to estimate the sleep problems among pregnant women during COVID-19 pandemic.

Methods

A systematic review was conducted through five academic databases. Relevant studies were extracted and their methodological quality was assessed using the Newcastle Ottawa Scale (NOS) checklist. Findings were synthesized using a meta-analysis approach. The report of present systematic review is part of a larger project registered in the PROSPERO International prospective register of systematic reviews (ID code: CRD42020181644)[27]. other related paper to this project is published elsewhere [25, 26]. This paper is prepared in accordance of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [28].

Search strategy

The five academic databases included Scopus, Medline/PubMed Central, ProQuest, ISI Web of Knowledge, and Embase which were searched systematically from December 2019 to July 2021. The search terms were extracted from PubMed Medical Subject Headings (MeSH). The main search terms were ‘sleep’, ‘COVID-19’ and ‘pregnancy’. The Boolean search method (AND/OR/NOT) was used to develop the search query. Search syntax was customized based on the advanced search attributes of each database. Key search components were selected based on PECO search strategy (i.e., Patient/Problem, Exposure, Comparison and Outcome) [29] to answer the research question. In the present study, key elements of exposure (Covid-19) and outcome (sleep problem) were selected. Search syntax is provided as supplementary file.

Inclusion criteria

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Observational studies were included if data on frequency or prevalence of sleep problems among pregnant women were reported. English, peer-reviewed papers published between December 2019 and July 2021 were included. There were no limitations regarding participant characteristics.

Outcomes

Primary outcome: Estimates of sleep problems frequency was the primary outcome. Sleep problems required assessments using valid and reliable psychometric scales, or confirmed with defined cut-off points. For instance, a global score of 5 or more in the Pittsburgh Sleep Quality Index or a total score of 8 or more in the Insomnia Severity Index indicate poor sleep quality [30], [31].

Secondary outcomes: Assessing the possible sources of heterogeneity and predictor variables of sleep problems prevalence among pregnant individuals.

Study screening and selection

Screening of title and abstract were done independently by two researchers based on the inclusion criteria and any disagreements were resolved by consensus. The full texts of potentially relevant studies were further examined based on the aforementioned criteria. In this process, relevant studies were selected.

Quality assessment

The Newcastle Ottawa Scale (NOS) was used to evaluate the methodological quality of the studies in observational studies. Three characteristics (i.e., selection, comparability, and outcome) are examined with the NOS checklist. The checklist evaluates methodological quality of cross-sectional studies based on seven items. The maximum quality score is 9 points and studies with less than 5 points are classified as having a high risk of bias [32]. No studies were excluded based on the quality rating. However, the impact of quality on pooled effect size was assessed via meta-regression.

Data extraction

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A pre-designed form was prepared to extract data from the studies included. The following items were extracted: first author's name, data collection dates, study design, country, number of participants, mean age, scales used to assess sleep problems, and numerical results regarding the frequency of sleep problems. It should also be noted that study selection, quality assessment, and data extraction were processes performed independently by two reviewers. Disagreements were resolved through discussion.

Data synthesis

A quantitative synthesis using STATA software version 14 was conducted. Meta-analysis was run using a random effects model because included studies were taken from different populations, and both within-study and between-study variances should be accounted for [33]. Severity of heterogeneity was estimated using the I^2 index. Heterogeneity is interpreted as (i) mild when $I^2$is less than 25%, (ii)moderate when I^2 is 25 to50%, (iii) severe when I^2 is 50 to75%, and (iv) highly severe when $I^2$is greater than 75%[34]. The key measure selected for the present study was prevalence of sleep problems. The numerical findings regarding prevalence of sleep problems were reported consistently in 6 studies, and are reported along with 95% confidence intervals (CI). To assess moderator analysis, meta-regression was carried out. Funnel plot and Eggers' Test were used to assess publication bias [35].Meta-trim with Fill and Trim method was used to correct probable publication bias [36]. Failure to correct the results in the presence of publication bias can lead to incorrect conclusions about the research question. Therefore, if the presence of publication bias is identified, the systematic reviewer should use the available methods to correct the results and present a more realistically justifiable conclusion. Although there are some methods recommended to correct publication bias [37], Fill and Trim method as a more conservative method was chosen and implemented. The Jackknife method was used for sensitivity analysis [38].

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Patient and Public Involvement: Patients or the public WERE NOT involved in the design, or conduct, or reporting, or dissemination plans of our research

Results

Study screening and selection process

The initial search in five databases resulted in 7,263 studies: Scopus (n=2,518), WOS (n=474), PubMed (n=338), Embase (n=1,426), and ProQuest (n=2,507). After removing duplicates, 5,647 papers retained based on title and abstract. Finally, 58 papers appeared to be potentially eligible and their full-texts were reviewed. In this process, 7 studies met the eligibility criteria, and were pooled in the meta-analysis. Figure 1 shows the search process based on the PRISMA flowchart.

Figure 1- PRISMA Flowchart of selected studies

Study description

A total of 7 papers with 2,808 participants from 4 different countries (Canada, Iran and Turkey (each 1 paper), and China (4 papers) were included. None of these papers gathered data during the national lockdown period in their respective countries. The smallest sample size was 45, and the largest sample size was 689, both from China. The mean age of participants was 30.66 years. All of the papers had cross-sectional design. Insomnia Severity Scale (ISI; n=4) and Pittsburgh Sleep Quality Index (PSQI; n=3) were used to assess sleep problems. Table 1 provides the summary characteristics of all included studies.

Quality assessment

All of the studies were categorized as being high quality studies. Recruitment of participants via online sampling and absence of estimate or justification regarding sample size were the most common problems encountered in the quality assessments.

Outcome measures

The pooled estimated prevalence of sleep problems was 56% [95% CI: 23 to 88%, I^2:99.81%, Tau^2:0.19]. Figure 2 provides the forest plot regarding the pooled prevalence.

Figure 2 - Forest plot regarding the pooled prevalence of sleep problems among pregnant women

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The probability of publication bias was assessed using Egger's test and funnel plot. Based on Egger's test (p=0.03) and funnel plot (Figure 3), publication bias emerged as probable.

Figure 3- Funnel plot assessing the publication bias among included studies

Due to the probability of publication bias, the fill-and-trim method was used to correct the estimated pooled measure. In this method, four studies were imputed, and the corrected results based on this method showed that the pooled prevalence of sleep problems was 13% (95% CI: 0 to 45%; p<.001). The resultant Funnel plot after trimming is provided in Figure 4.

Figure 4- Corrected Funnel plot based on fill and trim method

Furthermore, sensitivity analysis based on the Jack-knife method showed that the pooled effect size was not affected by a single study effect.

Figure 5- Assessment of small study effect based on Jack-knife method

Meta-regression (Table 2) showed that none of the examined variables explained the observed heterogeneity. Age was the only significant predictor of prevalence of sleep problems among pregnant women, and accounted for 64% of the variance. Each year increase in participants' age was associated with a 12% decrease in the prevalence of sleep problems during pregnancy.

Discussion

The main objective of this systematic review and meta-analysis was to examine the prevalence of sleep problems among pregnant women during the COVID-19 pandemic, and explore potential predictors of such sleep problems. To the best of our knowledge, the present study is the first to summarize the available evidence on sleep problems, the latter being exclusively determined using validated instruments. We found that the overall prevalence of sleep problems during pregnancy in COVID-19 pandemic was nominally 56%, and after bias estimate corrections, 13%. In a recent systematic review, the pooled prevalence of sleep problems in the general population during the COVID-19 pandemic was estimated at 18% (95% CI: 15-21%) [25]. In

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addition, Zhou et al.(2020) found that the prevalence of insomnia symptoms in pregnancy during the COVID-19 pandemic was 2.6% compared with 5.4% among non-pregnant women [11]. Interestingly, Xie et al.(2020) reported that 74.5% of women (not necessarily pregnant) complained about sleep problems in the COVID-19 pandemic [39]. In a current systematic review and meta-analysis, the corrected pooled estimated prevalence of sleep problems was 24% (95% CI: 19 to 29%) for female participants [26]. Thus, sleep problems are prevalent among pregnant women during the COVID-19 pandemic but not more than the prevalence found in the general population. Some factors may account for such observations, namely a decision to become pregnant is usually taken during a period of better mental health and more secure financial situation. Therefore, pregnan women may have improved and more stable mental health condition than non-pregnant women. Second, pregnant women receive the focus of family attention at all times, and such uniue support mechanisms may be especially implemented by family members during COVID-19 epidemic. Third, increased contact with medical workers for their prenatal care can provide support and decrease stress symptoms [40]. Therefore, these factors might lead to less insomnia in the pregnant women group.

Limitations

Among the limitations of the study we should point out that all studies were cross-sectional, absence of COVID-19-related assessments at the time of data collection, and the outcomes of the pregnancies among those with and without sleep problems in a consistent manner. Also, data regarding which trimester might be more susceptible to develop sleep disorders was not reported within the included studies. Consequently we could not compare the prevalence of sleep problems according to the pregnancy trimester. However, these results support the need for targeted studies in cohorts of pregnant women aimed at early detection of vulnerable sub-groups along with initial behavioral interventions aimed at mitigating the potential negative consequences of living through a pandemic on sleep during pregnancy and downstream outcomes.

Conclusion:

Pregnant women are likely to experience poor sleep quality when facing a pandemic such as COVID-19. Thus, special attention should be paid to pregnant women, and tools to identify those at risk for sleep problems along with effective interventions to prevent or mitigate the consequences of sleep problems during gestation are needed.

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Authors' contributions: ZA & AHP contributed to the conception, design of the study, and data collection. ZA & AHP contributed in data analysing and interpretation of data. ZA and FA drafted the manuscript. AHP & DG provided contributions to the literature review and discussion and substantially edited the primary manuscript and prepared the final version of the manuscript. All authors revised the manuscript, agreed to be fully accountable for ensuring the integrity and accuracy of the study, and read and approved the final version of the manuscript to be published. All the authors met the criteria for authorship, and they are listed as co-authors on the title page.

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Conflict of interests: None to declare

Ethics, approval, and consent to participate: The current research protocol was registered in PROSPERO International prospective register of systematic reviews (ID code: CRD42020181644.

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Clinical Trial Registry and Registration number: Not applicable. Current study is systematic review and registered in PROSPERO International prospective register of systematic reviews.

Data sharing statement: No additional unpublished data are available.

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Table 1- Summary of the characteristics of included studies

Authors	Year	Country	Collection Date	Sample Size	Mean Age (Years)	NOS	Sleep Problem Scale
Li [11]	2020	China	25 April to 9 May 2020	398		9	ISI
Khoury [41]	2021	Canada	June 3 and July 31, 2020	303	32.13	7	ISI
Xie [39]	2020	China	-	689	29.03	6	PSQI
Zhang [42]	2021	China	January to February 2020	456		6	PSQI
Zhou [11]	2020	China	February 28–March 12, 2020	859	33.25	9	ISI
Ahorsu [43]	2020	Iran	March 7 and April 21 2020	290	29.24	9	ISI
Alan [44]	2020	Turkey	25-30 April 2020	166	29.65	10	PSQI

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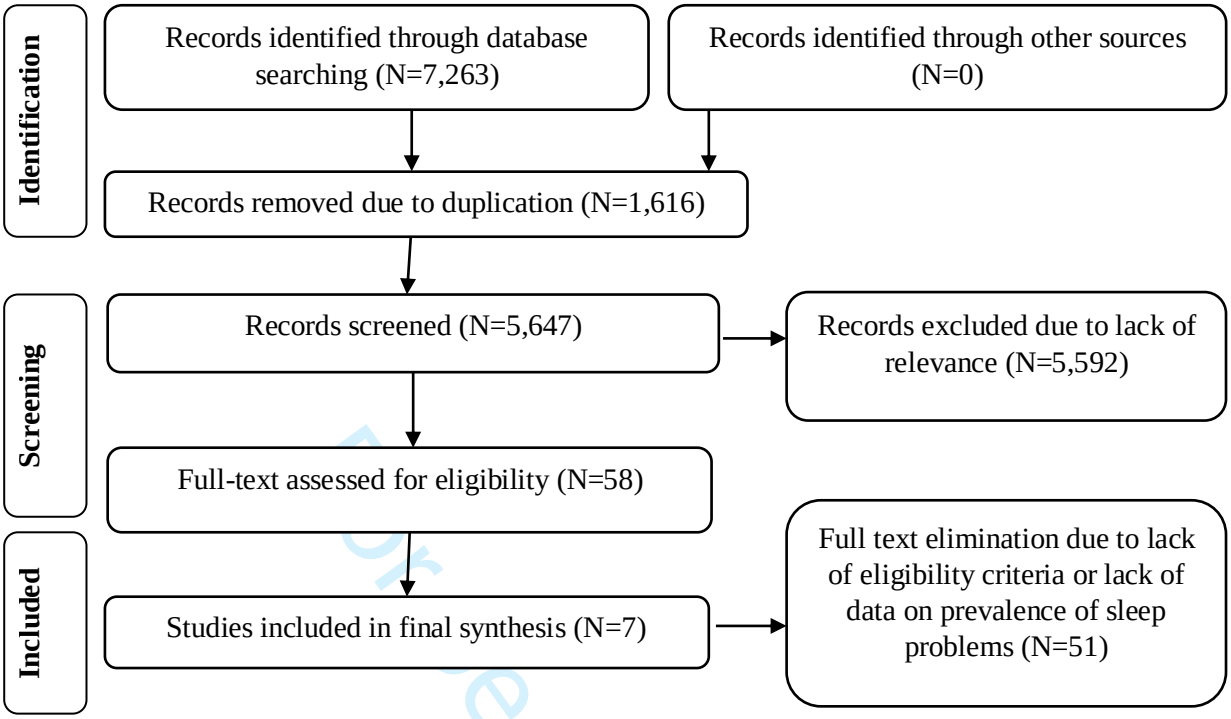
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Table 2. Results of meta-regression regarding estimated pooled prevalence						
univariable meta-regression						
Variable	Coefficient	S.E.	*p*	I^2 res. (%)	Adj. R^2 (%)	Tau^2
Country	0.15	0.08	0.11	99.72	32.23	0.05
Age	-0.15	0.05	0.07	98.76	64.01	0.04
NOS- score	0.02	0.06	0.74	99.82	-17.10	0.09
Measure of sleep	0.20	0.22	0.41	99.61	-3.06	0.09
multivariable meta-regression						
Country	0.08	0.09	0.42	98.61	63.91	0.04
Age	-0.12	0.06	0.17			

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Figure 1- PRISMA Flowchart of selected studies



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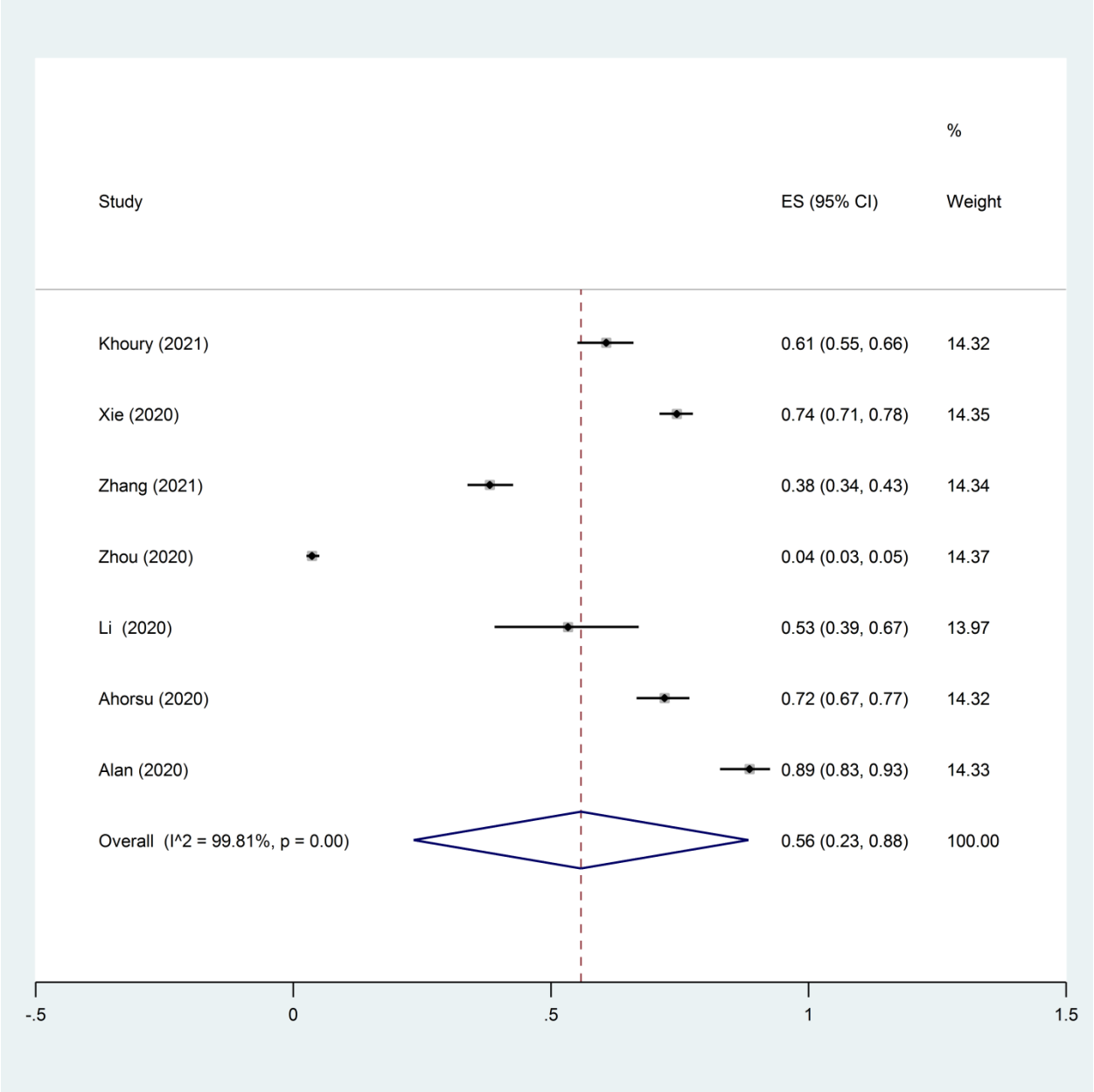


Figure 2 - Forest plot regarding the pooled prevalence of sleep problems among pregnant women

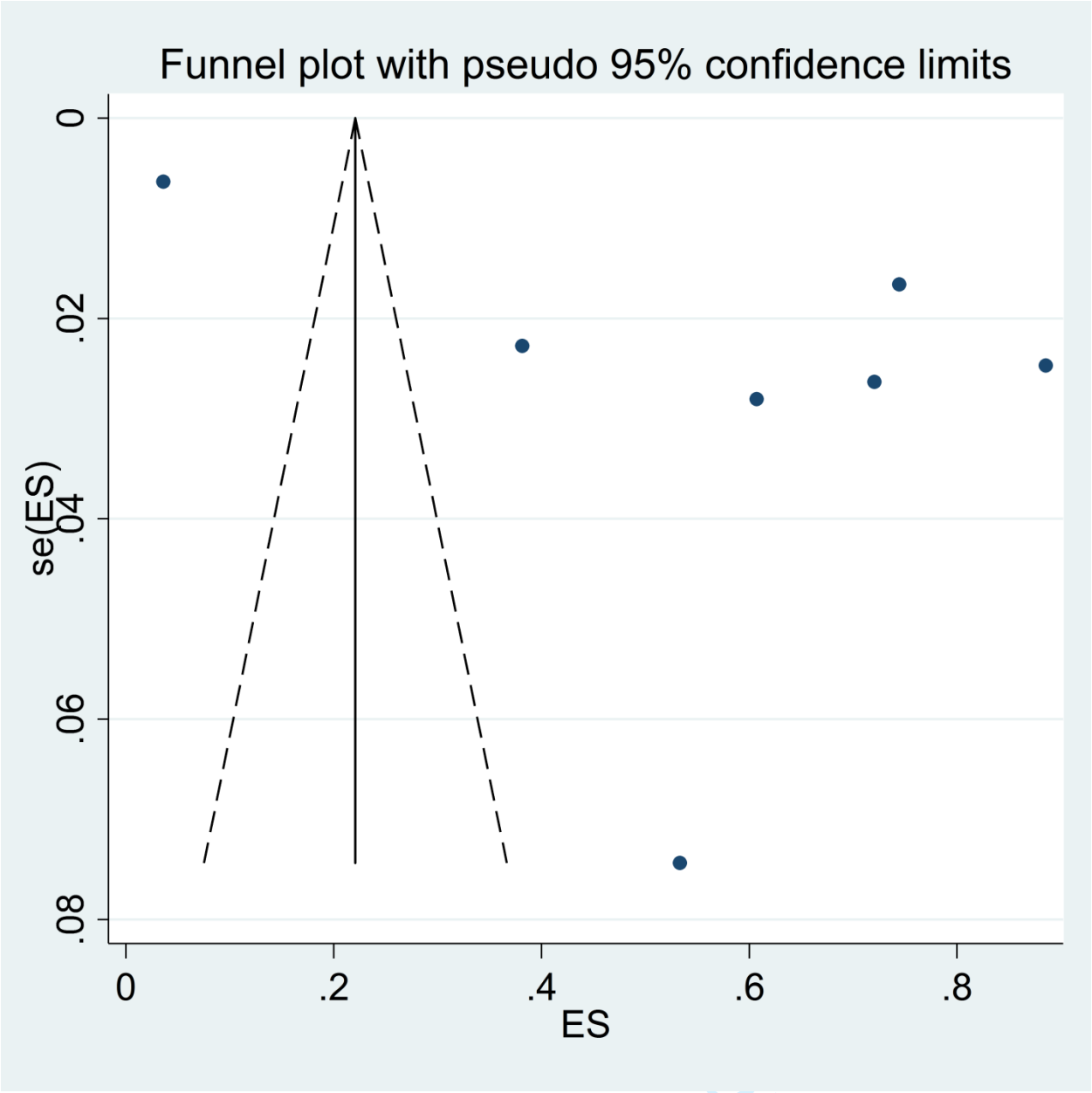


Figure 3- Funnel plot assessing the publication bias among included studies

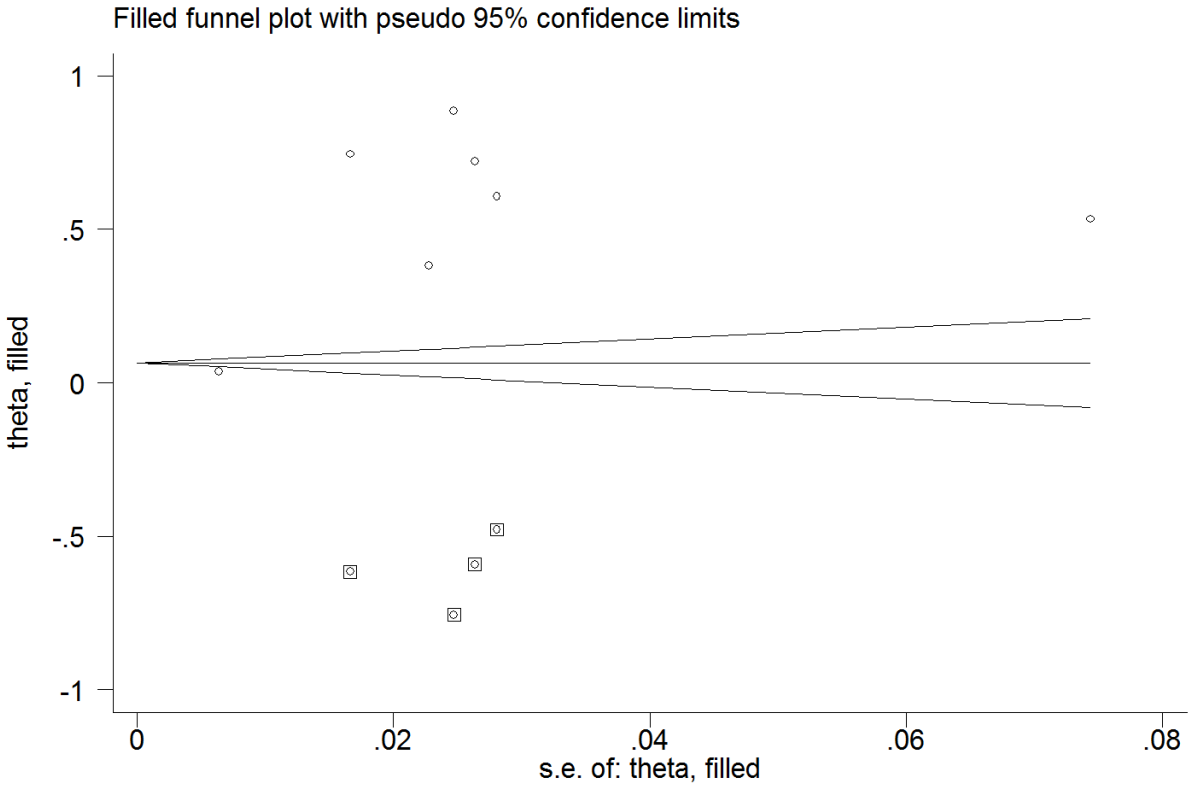


Figure 4- Corrected funnel plot based on fill and trim method

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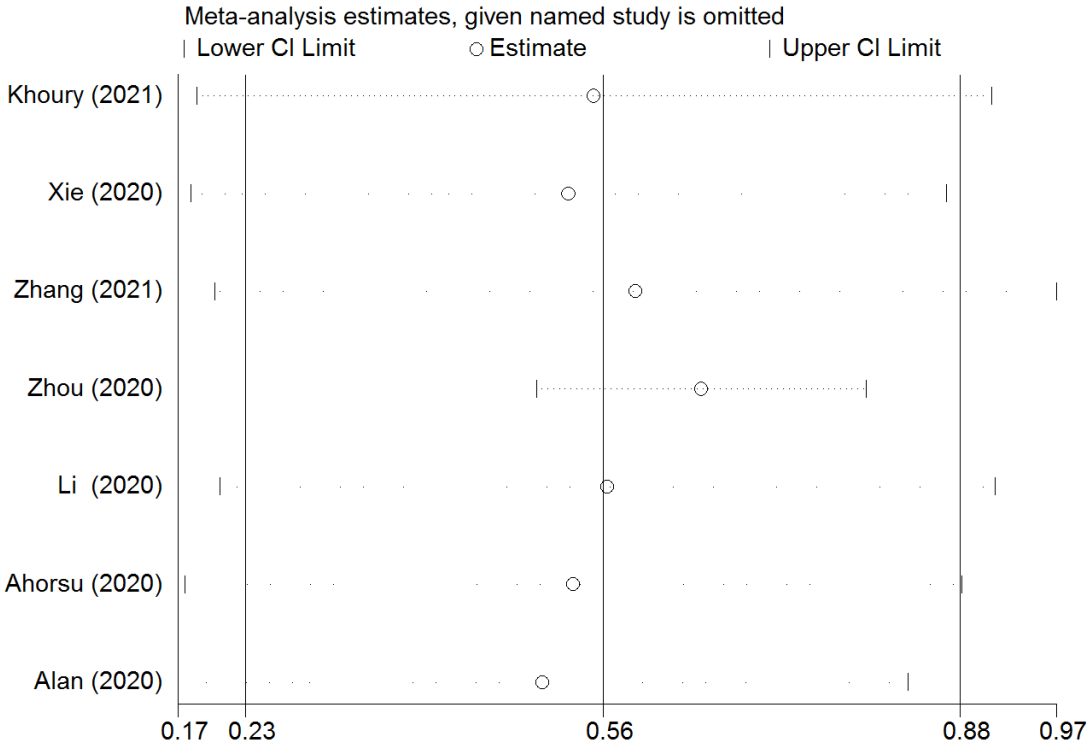


Figure 5- Assessment of small study effect based on Jack-knife method

view only

Database	Syntax
PubMed central	("2019 nCoV" OR 2019nCoV OR "2019 novel coronavirus" OR "COVID 19" OR COVID19 OR "new coronavirus" OR "novel coronavirus" OR "SARS CoV-2" OR (Wuhan AND coronavirus) OR "COVID 19" OR "SARS-CoV" OR "2019-nCoV" OR "SARS-CoV-2") AND (("insomnia" OR "sleep problems" OR "Sleep quality" OR Dyssomnia OR ("Sleep Disorders" AND Extrinsic) OR "Extrinsic Sleep Disorder" OR "Extrinsic Sleep Disorders" OR ("Sleep Disorder" AND Extrinsic) OR "Limit-Setting Sleep Disorder" OR "Limit Setting Sleep Disorder" OR "Limit-Setting Sleep Disorders" OR ("Sleep Disorders" AND "Limit-Setting") OR ("Sleep Disorder" AND Limit-Setting) OR ("Sleep Disorder" AND "Limit Setting") OR "Nocturnal Eating-Drinking Syndrome" OR ("Eating-Drinking Syndrome" AND Nocturnal) OR ("Eating-Drinking Syndromes" AND Nocturnal) OR "Nocturnal Eating Drinking Syndrome" OR "Nocturnal Eating-Drinking Syndromes" OR (Syndrome AND "Nocturnal Eating-Drinking") OR (Syndromes AND "Nocturnal Eating-Drinking") OR "Adjustment Sleep Disorder" OR "Adjustment Sleep Disorders" OR ("Sleep Disorders" AND Adjustment) OR ("Sleep Disorder" AND Adjustment) OR "Environmental Sleep Disorder" OR "Environmental Sleep Disorders" AND ("Sleep Disorders" AND Environmental) OR ("Sleep Disorder" AND Environmental)) AND ("2019/11/01"[PubDate] : "2020/12/31"[PubDate])
ProQuest	("2019 nCoV" OR 2019nCoV OR "2019 novel coronavirus" OR "COVID 19" OR COVID19 OR "new coronavirus" OR "novel coronavirus" OR "SARS CoV-2" OR (Wuhan AND coronavirus) OR "COVID 19" OR "SARS-CoV" OR "2019-nCoV" OR "SARS-CoV-2") AND (insomnia OR "sleep problems" OR "Sleep quality" OR Dyssomnia OR "Sleep Disorders" OR "Extrinsic Sleep Disorder" OR "Extrinsic Sleep Disorders" OR "Limit-Setting Sleep Disorder" OR "Limit Setting Sleep Disorder" OR "Limit-Setting Sleep Disorders" OR "Nocturnal Eating-Drinking Syndrome" OR ("Eating-Drinking Syndrome" AND Nocturnal) OR ("Eating-Drinking Syndromes" AND Nocturnal) OR "Nocturnal Eating Drinking Syndrome" OR "Nocturnal Eating-Drinking Syndromes" OR (Syndrome AND "Nocturnal Eating-Drinking") OR (Syndromes AND "Nocturnal Eating-Drinking") OR "Adjustment Sleep Disorder" OR "Adjustment Sleep Disorders" OR ("Sleep Disorders" AND Adjustment) OR ("Sleep Disorder" AND Adjustment) OR "Environmental Sleep Disorder" OR "Environmental Sleep Disorders" AND ("Sleep Disorders" AND Environmental)) OR ("Sleep Disorder" AND Environmental) AND ("2019/11/01"[PubDate] : "2020/12/31"[PubDate])
Scopus	("2019 nCoV" OR 2019ncov OR "2019 novel coronavirus" OR "COVID 19" OR covid19 OR "new coronavirus" OR "novel coronavirus" OR "SARS CoV-2" OR (wuhan AND coronavirus) OR "COVID 19" OR "SARS-CoV" OR "2019-nCoV" OR "SARS-CoV-2") AND (insomnia OR "sleep problems" OR "Sleep quality" OR dyssomnia OR "Sleep Disorders")
Embase	('2019 ncov' OR 2019ncov OR '2019 novel coronavirus'/exp OR '2019 novel coronavirus' OR covid19 OR 'new coronavirus' OR 'novel coronavirus' OR 'sars cov-2' OR (wuhan AND ('coronavirus'/exp OR coronavirus)) OR 'covid 19'/exp OR 'covid 19' OR 'sars-cov'/exp OR 'sars-cov' OR '2019-ncov' OR 'sars-cov-2') AND ('insomnia'/exp OR insomnia OR 'sleep problems' OR 'sleep quality'/exp OR 'sleep quality' OR 'dyssomnia'/exp OR dyssomnia OR 'sleep disorders'/exp OR 'sleep disorders')
ISI web of Knowledge	(TS= "2019 nCoV" OR TS= “2019ncov” OR TS= "2019 novel coronavirus" OR TS="COVID 19" OR TS=covid19 OR TS="new coronavirus" OR TS= "novel coronavirus" OR TS="SARS CoV-2" OR TS= (wuhan AND coronavirus) OR TS="COVID 19" OR TS="SARS-CoV" OR TS="2019-nCoV" OR TS="SARS-CoV-2") AND (TS=insomnia OR TS="sleep problems" OR TS="Sleep quality" OR TS= dyssomnia OR TS="Sleep Disorders")

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	1
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	2
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	3
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	3
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	3
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Appendix 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	4
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	4
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	4
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	4
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	4
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	5
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	5
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	NA
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	5
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	5
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	5
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	5
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	5
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	5

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	6
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	6
Study characteristics	17	Cite each included study and present its characteristics.	6
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	6
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table 1
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	6
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Figures 1-4 Table 2
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	7
	23b	Discuss any limitations of the evidence included in the review.	8
	23c	Discuss any limitations of the review processes used.	8
	23d	Discuss implications of the results for practice, policy, and future research.	9
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	3
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	3
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	10
Competing interests	26	Declare any competing interests of review authors.	10
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	10

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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Estimation of sleep problems among pregnant women during COVID-19 pandemic: A systematic review and meta-analysis

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Primary Subject Heading:	Mental health
Secondary Subject Heading:	Mental health, Obstetrics and gynaecology
Keywords:	SLEEP MEDICINE, OBSTETRICS, Maternal medicine < OBSTETRICS

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Systematic review Article

Estimation of sleep problems among pregnant women during COVID-19 pandemic: A systematic review and meta-analysis

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Abstract

Objective: *T*o estimate the sleep problems among pregnant women during the COVID-19 pandemic.

Eligibility criteria: English, peer-reviewed, observational studies published between December 2019 and July 2021 assessed and reported sleep problem prevalence using a valid and reliable measure were included

Information sources: Scopus, Medline/PubMed Central, ProQuest, ISI Web of Knowledge and Embase

Risk of bias assessment tool: The Newcastle Ottawa Scale (NOS) checklist

Synthesis of results: Prevalence of sleep problems was synthesis using STATA software version 14 using a random effects model. To assess moderator analysis, meta-regression was carried out. Funnel plot and Eggers' Test were used to assess publication bias. Meta-trim was used to correct probable publication bias. The Jackknife method was used for sensitivity analysis

Included studies: A total of seven cross sectional studies with 2,808 participants from four countries were included.

Synthesis of results: The pooled estimated prevalence of sleep problems was 56% [95% CI: 23 to 88%, I^2:99.81%, Tau^2:0.19]. Due to the probability of publication bias, the fill-and-trim method was used to correct the estimated pooled measure, which imputed four studies. The corrected results based on this method showed that pooled prevalence of sleep problems was 13% (95% CI: 0 to 45%; p<.001). Based on meta-regression, age was the only significant predictor of prevalence of sleep problems among pregnant women.

Limitations of evidence: all studies were cross-sectional, absence of assessment of sleep problems prior to COVID-19, and the outcomes of the pregnancies among those with and without sleep problems in a consistent manner are among limitation of current review.

Interpretation: Pregnant women have experienced significant declines in sleep quality when faced with the COVID-19 pandemic. The short-term and long-term implications of such alterations in sleep on gestational and offspring outcomes are unclear and warrant further studies.

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Keywords: COVID-19; Sleep problems; Pregnancy; meta-analysis

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Article summary

Strengths of this study

- Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guideline was used to report the findings.
- Newcastle Ottawa (NOS) checklist was used to assess methodological quality of included studies.
- Five academic databases were systematically searched to increase the comprehensiveness of search.

limitations of this study

- All retrieved studies were cross-sectional.
- The outcomes of the pregnancies among those with and without sleep problems was not assessed in included studies.

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Introduction

Pregnant women are exposed to additional psychological stress due to the indirect adverse effects of COVID-19 pandemic [1, 2]. The unintended consequences of COVID-19 can negatively impact the health of pregnant women and raise substantial concerns for this population [3].

Among these, mental health and psychological consequences, as well as sleep problems have become ubiquitously manifest across the world [4-7]. On the other hand, people who have a good quality of sleep have fewer post-traumatic stress disorder (PTSD) symptoms associated with the COVID-19 outbreak [8, 9].The United Nations has reported the COVID-19 pandemic as "the seeds of a major mental health crisis around the globe"[10]. In this context, pregnant women may often develop psychiatric symptoms, reduced sleep quality, which may enhance their vulnerability relative to others to the pandemic and its consequences and may be more at risk of COVID-19 infection and worse outcomes [11, 12]. Outbreaks of infectious diseases when combined with pregnancy are associated with major psychologic distress and significant symptoms, including poor sleep quality [13].

It is now well established that pregnancy in general is fraught with a higher risk of developing insomnia complaints, emergence of excessive daytime sleepiness and reports of decreased subjective sleep quality [14, 15]. Available studies estimated the prevalence of sleep problems such as insomnia and frequent awakenings in pregnant women at 46–78%, with the quality of sleep declining towards the third trimester[6]. Sleep disorders are associated with various adverse pregnancy-related health outcomes, including poorer fetal outcomes, specifically birthweight, growth, preterm birth and stillbirth [16]. Additionally, the occurrence of insomnia during pregnancy may increase the risk of postpartum depression [17].The result of a meta-analysis showed that the prevalence of anxiety and depression in pregnant women during COVID-19 pandemic was 15.8 and 25.8%, respectively [1].

Pregnant women may be more likely to develop anxiety, the latter being powerfully linked with poor sleep, particularly during the COVID-19 pandemic [18]. The apparent associations between pregnancy and sleep disturbances, as well as the aforementioned adverse outcomes of COVID-19 in pregnant women raise the strong possibility that the changes in lifestyle and many other environmental circumstances imposed by the COVID-19 crisis may translate into enhanced repercussions on sleep during pregnancy. As such, we would anticipate that the prevalence of insomnia, a disorder associated with increased risk of chronic illnesses, poor mental health and

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functional limitations might be further enhanced [19]. It would be reasonable to assume that pregnant women may experience incremental sleep problems due to the social distancing and isolation required during the COVID-19 pandemic [20]. Thus, early diagnosis and timely treatment of such sleep problems may mitigate the risk for adverse gestational and perinatal outcomes [21]. Despite the impact of the COVID-19 on sleep quality and quantity [22-26], there is no systematic review or meta-analysis has yet been conducted to examine the impact of the pandemic on the prevalence of sleep problems during pregnancy. Therefore, this systematic review and meta-analysis was conducted to estimate the sleep problems among pregnant women during COVID-19 pandemic.

Methods

A systematic review was conducted through five academic databases. Relevant studies were extracted and their methodological quality was assessed using the Newcastle Ottawa Scale (NOS) checklist. Findings were synthesized using a meta-analysis approach. The report of present systematic review is part of a larger project registered in the PROSPERO International prospective register of systematic reviews (ID code: CRD42020181644)[27]. other related paper to this project is published elsewhere [25, 26]. This paper is prepared in accordance of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [28].

Search strategy

The five academic databases included Scopus, Medline/PubMed Central, ProQuest, ISI Web of Knowledge, and Embase which were searched systematically from December 2019 to July 2021. The search terms were extracted from PubMed Medical Subject Headings (MeSH). The main search terms were ‘sleep’, ‘COVID-19’ and ‘pregnancy’. The Boolean search method (AND/OR/NOT) was used to develop the search query. Search syntax was customized based on the advanced search attributes of each database. Key search components were selected based on PECO search strategy (i.e., Patient/Problem, Exposure, Comparison and Outcome) [29] to answer the research question. In the present study, key elements of exposure (Covid-19) and outcome (sleep problem) were selected.

Inclusion criteria

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Observational studies were included if data on frequency or prevalence of sleep problems among pregnant women were reported. English, peer-reviewed papers published between December 2019 and July 2021 were included. There were no limitations regarding participant characteristics.

Outcomes

Primary outcome: Estimates of sleep problems frequency was the primary outcome. Sleep problems required assessments using valid and reliable psychometric scales, or confirmed with defined cut-off points. For instance, a global score of 5 or more in the Pittsburgh Sleep Quality Index or a total score of 8 or more in the Insomnia Severity Index indicate poor sleep quality [30], [31].

Secondary outcomes: Assessing the possible sources of heterogeneity and predictor variables of sleep problems prevalence among pregnant individuals.

Study screening and selection

Screening of title and abstract were done independently by two researchers based on the inclusion criteria and any disagreements were resolved by consensus. The full texts of potentially relevant studies were further examined based on the aforementioned criteria. In this process, relevant studies were selected.

Quality assessment

The Newcastle Ottawa Scale (NOS) was used to evaluate the methodological quality of the studies in observational studies. Three characteristics (i.e., selection, comparability, and outcome) are examined with the NOS checklist. The checklist evaluates methodological quality of cross-sectional studies based on seven items. The maximum quality score is 9 points and studies with less than 5 points are classified as having a high risk of bias [32]. No studies were excluded based on the quality rating. However, the impact of quality on pooled effect size was assessed via meta-regression.

Data extraction

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A pre-designed form was prepared to extract data from the studies included. The following items were extracted: first author's name, data collection dates, study design, country, number of participants, mean age, scales used to assess sleep problems, and numerical results regarding the frequency of sleep problems. It should also be noted that study selection, quality assessment, and data extraction were processes performed independently by two reviewers. Disagreements were resolved through discussion.

Data synthesis

A quantitative synthesis using STATA software version 14 was conducted. Meta-analysis was run using a random effects model because included studies were taken from different populations, and both within-study and between-study variances should be accounted for [33]. Severity of heterogeneity was estimated using the I^2 index. Heterogeneity is interpreted as (i) mild when $I^2$is less than 25%, (ii)moderate when I^2 is 25 to50%, (iii) severe when I^2 is 50 to75%, and (iv) highly severe when $I^2$is greater than 75%[34]. The key measure selected for the present study was prevalence of sleep problems. The numerical findings regarding prevalence of sleep problems were reported consistently in 6 studies, and are reported along with 95% confidence intervals (CI). To assess moderator analysis, meta-regression was carried out. Funnel plot and Eggers' Test were used to assess publication bias [35].Meta-trim with Fill and Trim method was used to correct probable publication bias [36]. Failure to correct the results in the presence of publication bias can lead to incorrect conclusions about the research question. Therefore, if the presence of publication bias is identified, the systematic reviewer should use the available methods to correct the results and present a more realistically justifiable conclusion. Although there are some methods recommended to correct publication bias [37], Fill and Trim method as a more conservative method was chosen and implemented. The Jackknife method was used for sensitivity analysis [38].

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Patient and Public Involvement: Patients or the public WERE NOT involved in the design, or conduct, or reporting, or dissemination plans of our research

Results

Study screening and selection process

The initial search in five databases resulted in 7,263 studies: Scopus (n=2,518), WOS (n=474), PubMed (n=338), Embase (n=1,426), and ProQuest (n=2,507). After removing duplicates, 5,647 papers retained based on title and abstract. Finally, 58 papers appeared to be potentially eligible and their full-texts were reviewed. In this process, 7 studies met the eligibility criteria, and were pooled in the meta-analysis. Figure 1 shows the search process based on the PRISMA flowchart.

Figure 1- PRISMA Flowchart of selected studies

Study description

A total of 7 papers with 2,808 participants from 4 different countries (Canada, Iran and Turkey (each 1 paper), and China (4 papers) were included. None of these papers gathered data during the national lockdown period in their respective countries. The smallest sample size was 45, and the largest sample size was 689, both from China. The mean age of participants was 30.66 years. All of the papers had cross-sectional design. Insomnia Severity Scale (ISI; n=4) and Pittsburgh Sleep Quality Index (PSQI; n=3) were used to assess sleep problems. Table 1 provides the summary characteristics of all included studies.

Quality assessment

All of the studies were categorized as being high quality studies. Recruitment of participants via online sampling and absence of estimate or justification regarding sample size were the most common problems encountered in the quality assessments.

Outcome measures

The pooled estimated prevalence of sleep problems was 56% [95% CI: 23 to 88%, I^2:99.81%, Tau^2:0.19]. Figure 2 provides the forest plot regarding the pooled prevalence.

Figure 2 - Forest plot regarding the pooled prevalence of sleep problems among pregnant women

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The probability of publication bias was assessed using Egger's test and funnel plot. Based on Egger's test (p=0.03) and funnel plot (Figure 3), publication bias emerged as probable.

Figure 3- Funnel plot assessing the publication bias among included studies

Due to the probability of publication bias, the fill-and-trim method was used to correct the estimated pooled measure. In this method, four studies were imputed, and the corrected results based on this method showed that the pooled prevalence of sleep problems was 13% (95% CI: 0 to 45%; p<.001). The resultant Funnel plot after trimming is provided in Figure 4.

Figure 4- Corrected Funnel plot based on fill and trim method

Furthermore, sensitivity analysis based on the Jack-knife method showed that the pooled effect size was not affected by a single study effect (Figure 5).

Figure 5- Assessment of small study effect based on Jack-knife method

Meta-regression (Table 2) showed that none of the examined variables explained the observed heterogeneity. Age was the only significant predictor of prevalence of sleep problems among pregnant women, and accounted for 64% of the variance. Each year increase in participants' age was associated with a 12% decrease in the prevalence of sleep problems during pregnancy.

Discussion

The main objective of this systematic review and meta-analysis was to examine the prevalence of sleep problems among pregnant women during the COVID-19 pandemic, and explore potential predictors of such sleep problems. To the best of our knowledge, the present study is the first to summarize the available evidence on sleep problems, the latter being exclusively determined using validated instruments. We found that the overall prevalence of sleep problems during pregnancy in COVID-19 pandemic was nominally 56%, and after bias estimate corrections, 13%. In a recent systematic review, the pooled prevalence of sleep problems in the general population during the COVID-19 pandemic was estimated at 18% (95% CI: 15-21%) [25]. In

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addition, Zhou et al.(2020) found that the prevalence of insomnia symptoms in pregnancy during the COVID-19 pandemic was 2.6% compared with 5.4% among non-pregnant women [11]. Interestingly, Xie et al.(2020) reported that sleep problems among pregnant women were 74.5% during the COVID-19 pandemic vs. 69.1% before COVID-19 pandemic [39]. There may be possible reasons for this discrepancy. The study of Xie et al. was conducted at the beginning of the pandemic, while the study of Zhou et al. was pursued later. Numerous studies have shown that fear and mental health problems were more prominent at the beginning of the pandemic. Since these studies employed different tools to evaluate sleep problems, it is also likely that the differences in findings may be explained by such fact. In a current systematic review and meta-analysis, the corrected pooled estimated prevalence of sleep problems was 24% (95% CI: 19 to 29%) for female participants [26]. Thus, sleep problems are prevalent among pregnant women during the COVID-19 pandemic, but not more than the prevalence found in the general population. Some factors may account for such observations, namely a decision to become pregnant is usually taken during a period of better mental health and more secure financial situation. Therefore, pregnant women may have improved and more stable mental health condition than non-pregnant women. Second, pregnant women receive the focus of family attention at all times, and such unique support mechanisms may be especially implemented by family members during COVID-19 epidemic. Third, increased contact with medical workers for their prenatal care can provide support and decrease stress symptoms [40]. Therefore, these factors might lead to less insomnia in the pregnant women group.

Limitations

Among the limitations of the study we should point out that all studies were cross-sectional, absence of COVID-19-related assessments at the time of data collection, and the outcomes of the pregnancies among those with and without sleep problems in a consistent manner. Also, data regarding which trimester might be more susceptible to develop sleep disorders was not reported within the included studies. Consequently we could not compare the prevalence of sleep problems according to the pregnancy trimester. However, these results support the need for targeted studies in cohorts of pregnant women aimed at early detection of vulnerable sub-groups along with initial behavioral interventions aimed at mitigating the potential negative consequences of living through a pandemic on sleep during pregnancy and downstream outcomes.

Conclusion:

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Pregnant women are likely to experience poor sleep quality when facing a pandemic such as COVID-19. Thus, special attention should be paid to pregnant women, and tools to identify those at risk for sleep problems along with effective interventions to prevent or mitigate the consequences of sleep problems during gestation are needed.

Authors' contributions: ZA & AHP contributed to the conception, design of the study, and data collection. ZA & AHP contributed in data analysing and interpretation of data. ZA and FA drafted the manuscript. AHP & DG provided contributions to the literature review and discussion and substantially edited the primary manuscript and prepared the final version of the manuscript. All authors revised the manuscript, agreed to be fully accountable for ensuring the integrity and accuracy of the study, and read and approved the final version of the manuscript to be published. All the authors met the criteria for authorship, and they are listed as co-authors on the title page.

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Conflict of interests: None to declare

Ethics, approval, and consent to participate: The current research protocol was registered in PROSPERO International prospective register of systematic reviews (ID code: CRD42020181644.

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Clinical Trial Registry and Registration number: Not applicable. Current study is systematic review and registered in PROSPERO International prospective register of systematic reviews.

Data sharing statement: No additional unpublished data are available.

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Table 1- Summary of the characteristics of included studies

Authors	Year	Country	Collection Date	Sample Size	Mean Age (Years)	NOS	Sleep Problem Scale
Li [11]	2020	China	25 April to 9 May 2020	398		9	ISI
Khoury [41]	2021	Canada	June 3 and July 31, 2020	303	32.13	7	ISI
Xie [39]	2020	China	-	689	29.03	6	PSQI
Zhang [42]	2021	China	January to February 2020	456		6	PSQI
Zhou [11]	2020	China	February 28–March 12, 2020	859	33.25	9	ISI
Ahorsu [43]	2020	Iran	March 7 and April 21 2020	290	29.24	9	ISI
Alan [44]	2020	Turkey	25-30 April 2020	166	29.65	10	PSQI

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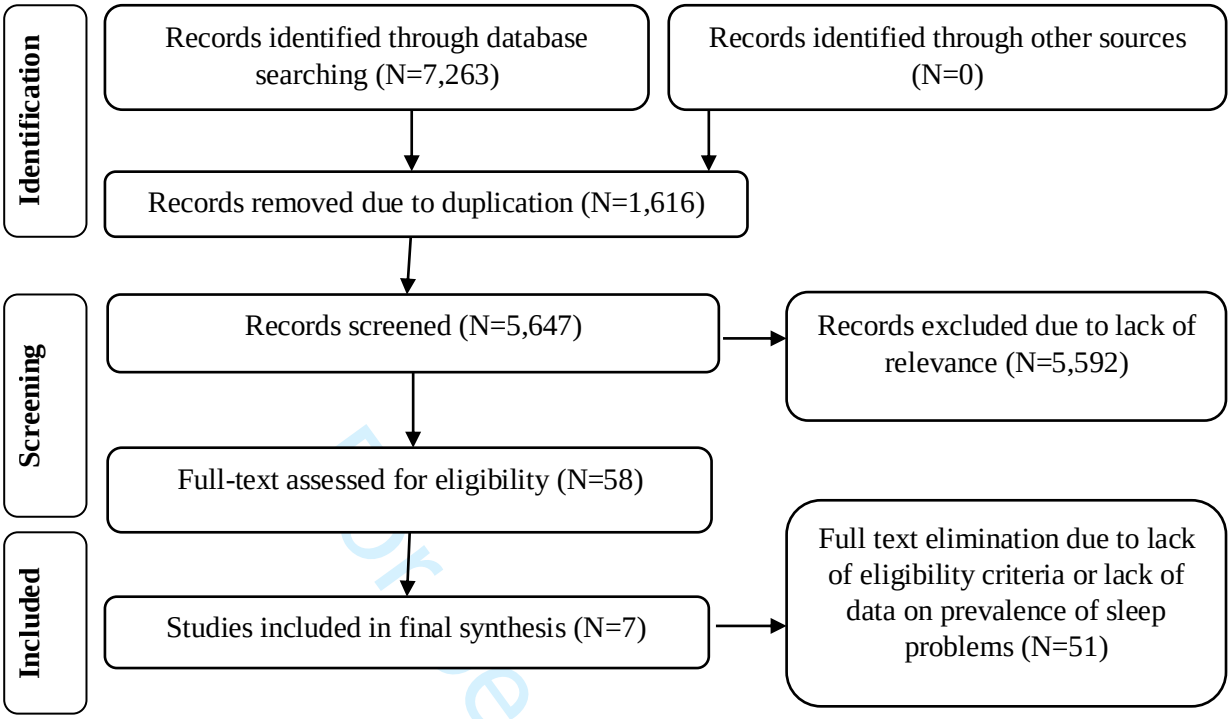
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Table 2. Results of meta-regression regarding estimated pooled prevalence						
univariable meta-regression						
Variable	Coefficient	S.E.	*p*	I^2 res. (%)	Adj. R^2 (%)	Tau^2
Country	0.15	0.08	0.11	99.72	32.23	0.05
Age	-0.15	0.05	0.07	98.76	64.01	0.04
NOS- score	0.02	0.06	0.74	99.82	-17.10	0.09
Measure of sleep	0.20	0.22	0.41	99.61	-3.06	0.09
multivariable meta-regression						
Country	0.08	0.09	0.42	98.61	63.91	0.04
Age	-0.12	0.06	0.17			

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Figure 1- PRISMA Flowchart of selected studies



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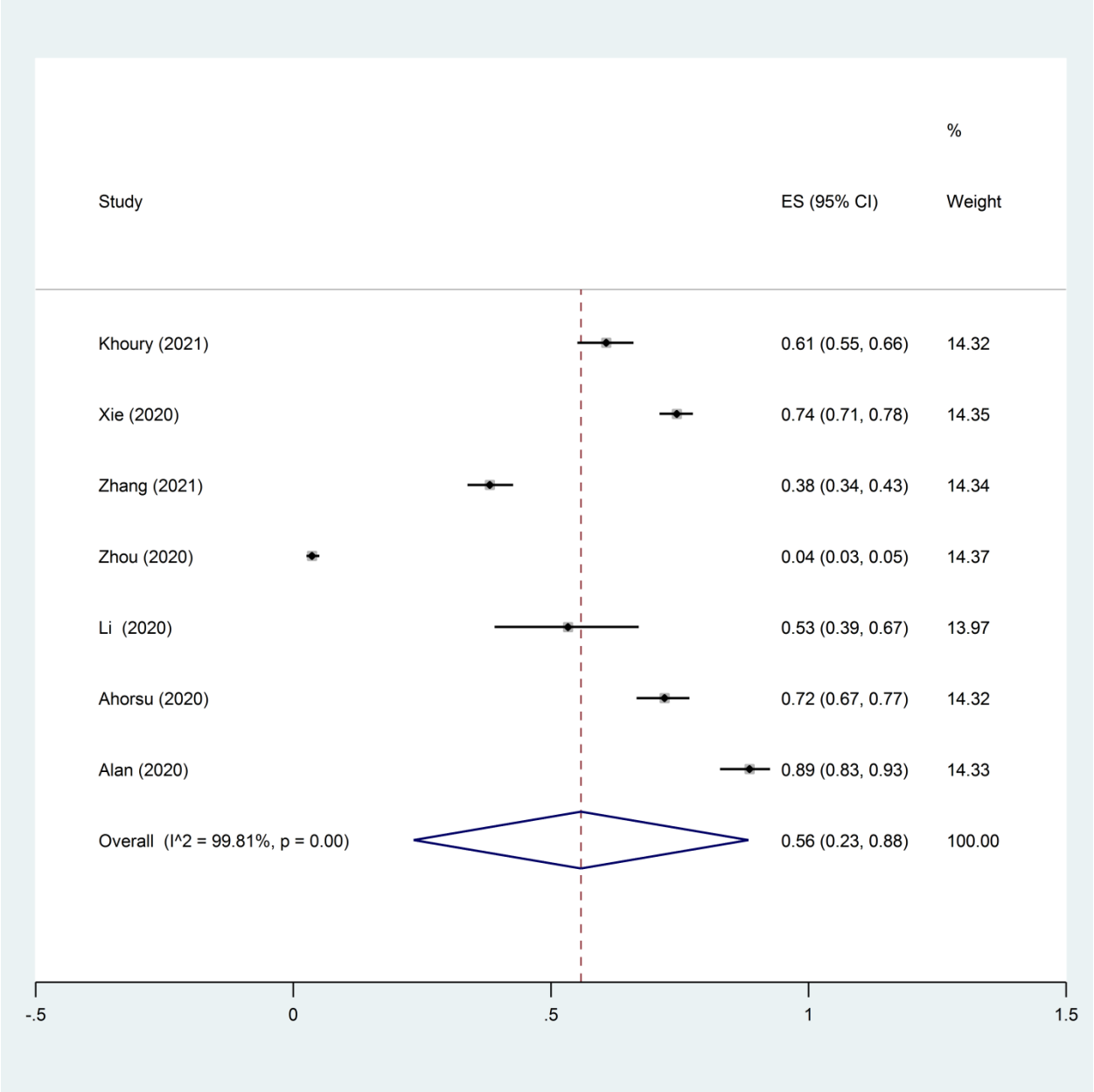


Figure 2 - Forest plot regarding the pooled prevalence of sleep problems among pregnant women

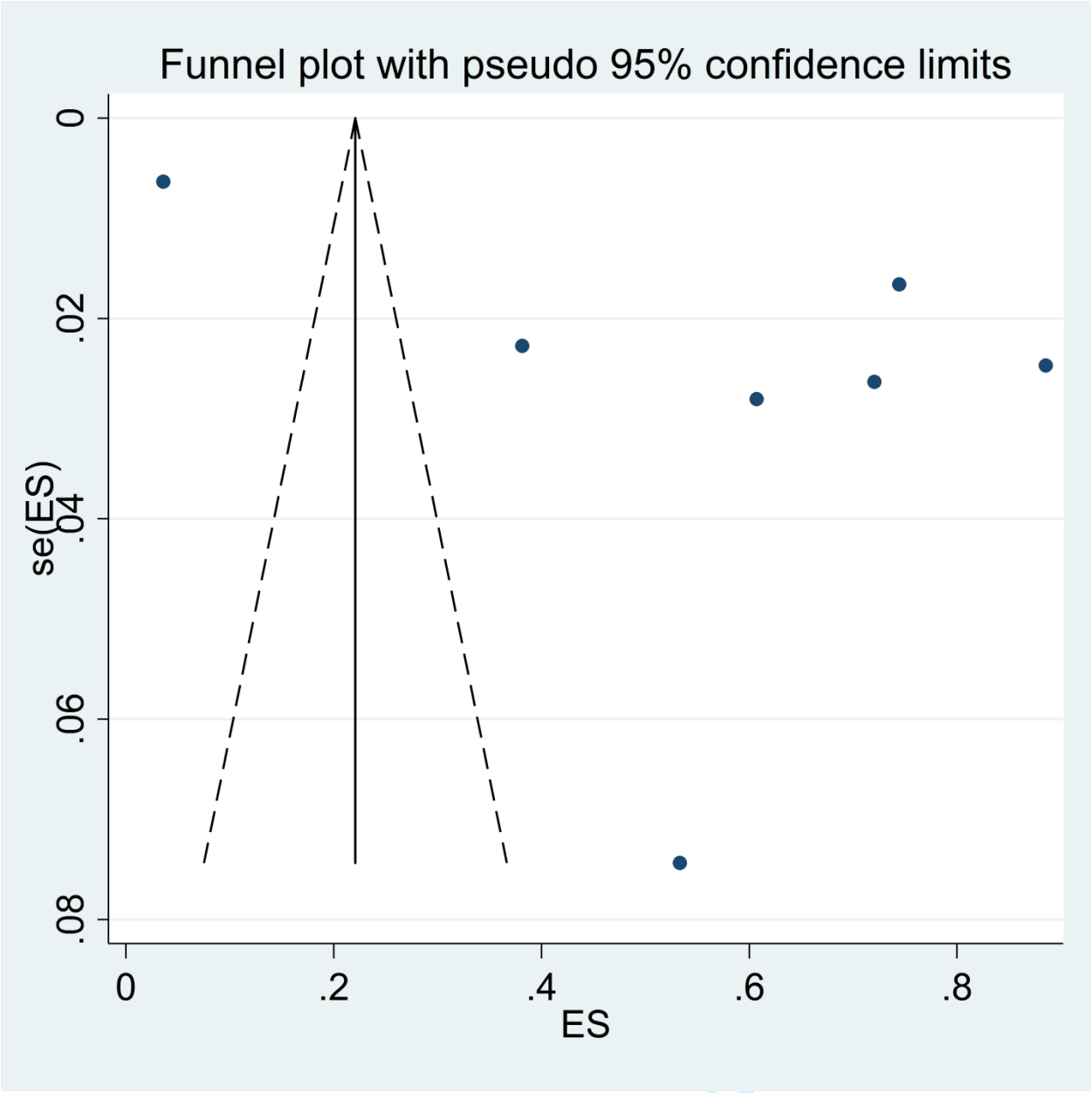


Figure 3- Funnel plot assessing the publication bias among included studies

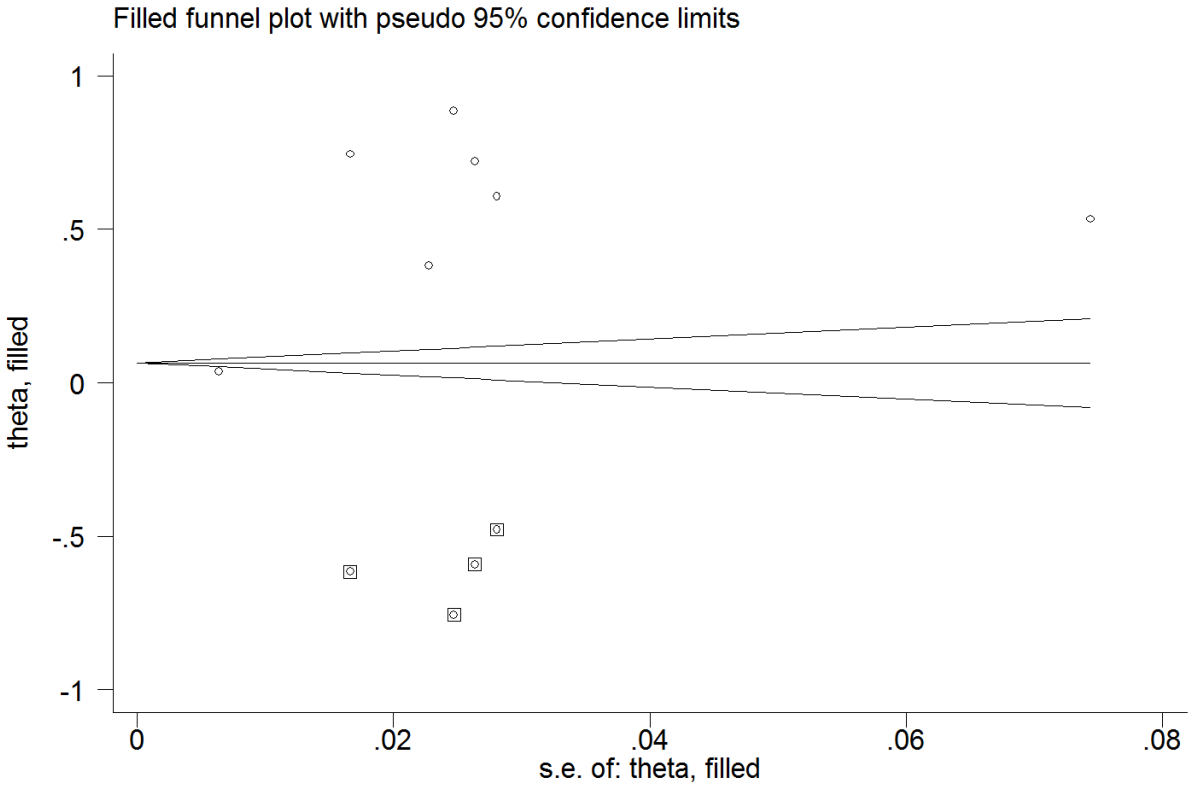


Figure 4- Corrected funnel plot based on fill and trim method

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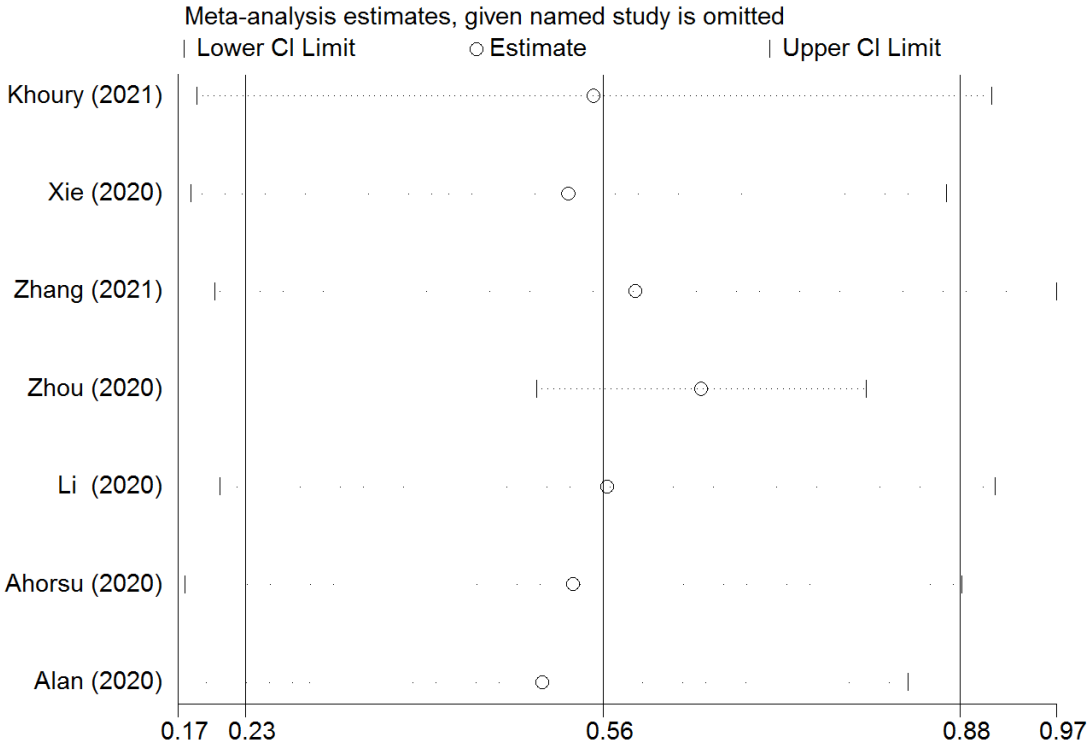


Figure 5- Assessment of small study effect based on Jack-knife method

view only

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	1
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	2
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	3
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	3
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	3
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Appendix 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	4
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	4
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	4
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	4
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	4
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	5
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	5
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	NA
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	5
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	5
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	5
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	5
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	5
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	5

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	6
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	6
Study characteristics	17	Cite each included study and present its characteristics.	6
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	6
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table 1
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	6
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Figures 1-4 Table 2
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	7
	23b	Discuss any limitations of the evidence included in the review.	8
	23c	Discuss any limitations of the review processes used.	8
	23d	Discuss implications of the results for practice, policy, and future research.	9
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	3
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	3
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	10
Competing interests	26	Declare any competing interests of review authors.	10
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	10

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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